



Elevate Requirements Management to the Next Level



Bob Parro, VP Product Development

3/20/2014

- Who is River North Solutions
- Systems engineer context
- Importance of requirements management (RM)
- RM tool – DOORS
- Day in the Life of RM document
- Summary

River North Profile

IBM Software Expertise

Requirements Management

- Rational DOORS
- Rational DOORS Next Generation
- Rational Requirements Composer
- Rational Requisite Pro

Collaborative Lifecycle Management

- Rational Team Concert (RTC)
- Rational Quality Manager (RQM)
- Jazz platform

Document Publishing

- Rational Publishing Engine (RPE)

Modeling

- Rhapsody (Systems & Software)
- Rational Systems Architect (EA)
- Rational Software Architect (IT)

Product/Portfolio Management

- Rational Focal Point

Training

- All above IBM products
- Systems Engineering methodologies

River North Rational Add-on Products

- *Ambiguous Words Detector*™ (AWD)
- *ArrangeMe*™ – DOORS data manipulation
- *Requirement Metrics Dashboard*™ (RMD)
- *ChangeTracker*™ – revision rich text markup
- *Export Me*™ – automated doc publishing
- *Allocated Requirements Tool*™ (ART)

Industry Experience

- Energy
- Medical Device
- Transportation
- Aerospace & Defense
- Telecommunications
- Healthcare/Providers

Clients (partial list)

- NuScale Power
- Westinghouse Nuclear
- Baxter Healthcare
- Hospira
- Bayer Healthcare
- Eli Lilly
- Roche Diagnostics
- Ansaldo STS
- Crane Aerospace
- Northrop Grumman
- Comcast
- Multiple Healthcare systems in Midwest, Southwest, West Coast

Professional Society Leadership

INCOSE

- Risk Management WG Co-Chair
- Midwest Biomedical Healthcare WG
- Chicagoland Chapter President

IEEE

- Board of Directors

Clinical Information Systems

- Selection, implementation, conversion

Key River North Smarter Solutions

Requirements Management Accelerator

- Data input and manipulation automation
- Requirements quality automation
- Product variant productivity
- Process automation
- Data/document publishing automation
- Metrics

Staff Augmentation

- Systems engineers

Key River North Partners

- Sodius
- GEBS Reporting
- Island Training

Executive Management

- Mike Parro, Principal
- Bob Parro, VP Business Development & Product Development

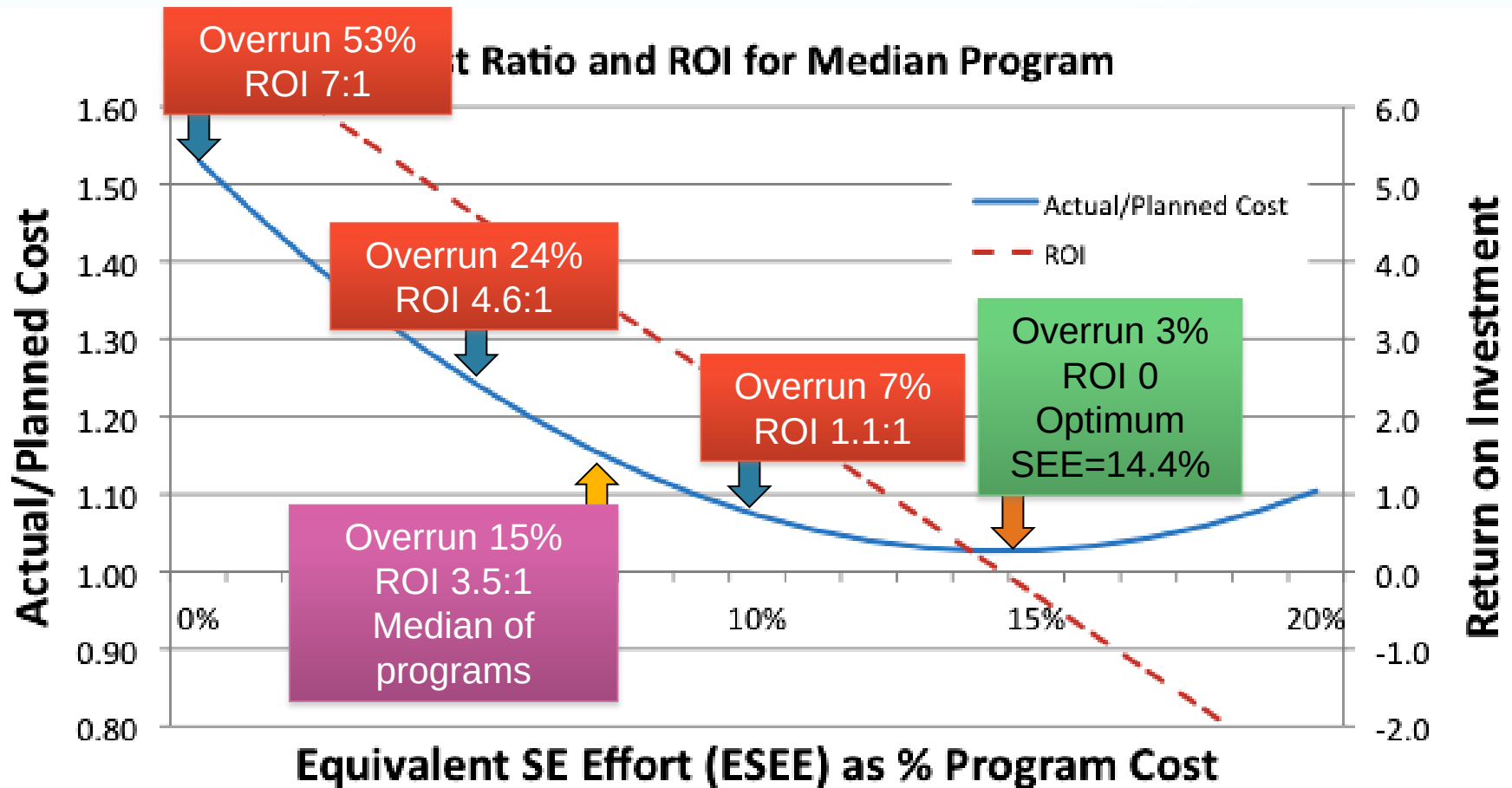
Contact Information

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info@rivernorthsolutions.com



Accelerate Requirements Management

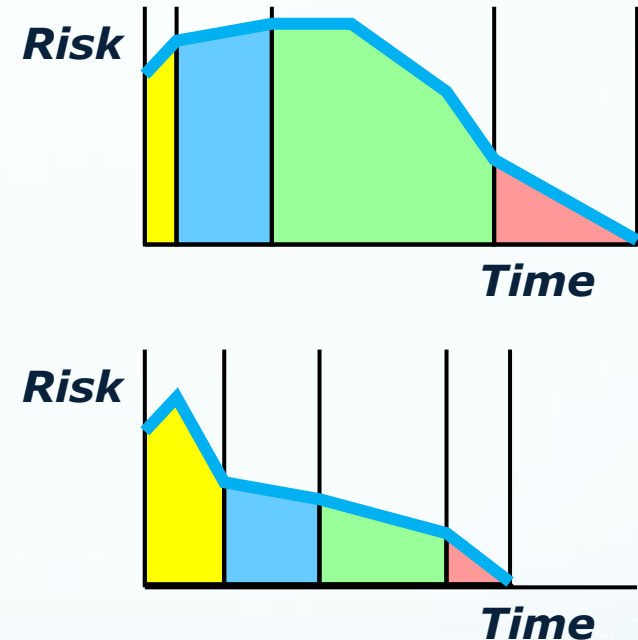
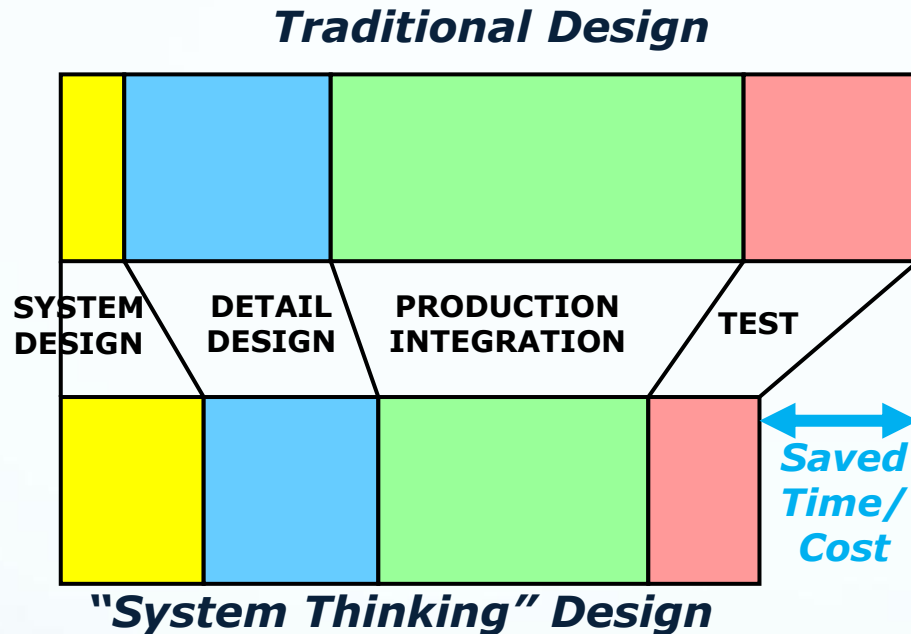
SE Spend for Optimum ROI



Honour, EC, *Systems Engineering Return on Investment*,
PhD thesis, Univ South Australia 2013

<http://www.hcode.com/seroi/index.html>

Accelerate Requirements Management Systems Engineering Impact



- Better systems engineering leads to
 - Lower cost
 - Shorter schedule
 - Better system value
- Eric Honour
 - INCOSE Founder
 - INCOSE Fellow
 - Former President

Accelerate Requirements Management

Gartner Nov 2013 ALM Market Analysis

- ALM – Application Lifecycle Management...
- Software requirements definition & management
- Software change & configuration management
- Software project planning, with a current focus on agile planning
- Work item management
- Quality management, including defect management
- <http://www.gartner.com/technology/reprints.do?id=1-1NHDUJ3&ct=131126&st=sb>

Figure 1. Magic Quadrant for Application Development Life Cycle Management



Poor Requirements Management has a Significant Impact on your Business

Requirements Rework

- Errors, late detected in the Maintenance phase can cost up to **200 times** more than detected early in Requirement Analysis phase¹
- More than 40%** of development budget can be consumed by poor requirements²

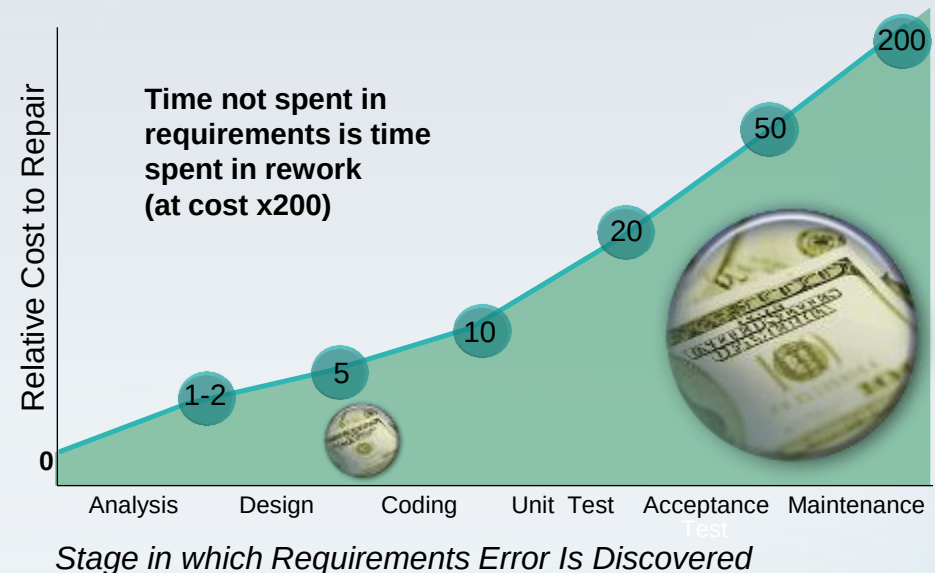
Requirements issues drive excessive rework, delays, poor quality, and project failures

Project Impacts

- 41%** of projects fail to deliver the expected business value and ROI³
- 49%** of projects overrun original estimates³
- 28%** of projects on time and on budget⁴

Project Delays

- Being late to market by 6 months or more will cost organizations 33% of the 5-year ROI⁵



“Our research indicates 80-plus percent of development failures result directly from poor requirements gathering, management, and analysis.”

IDC, November 2007

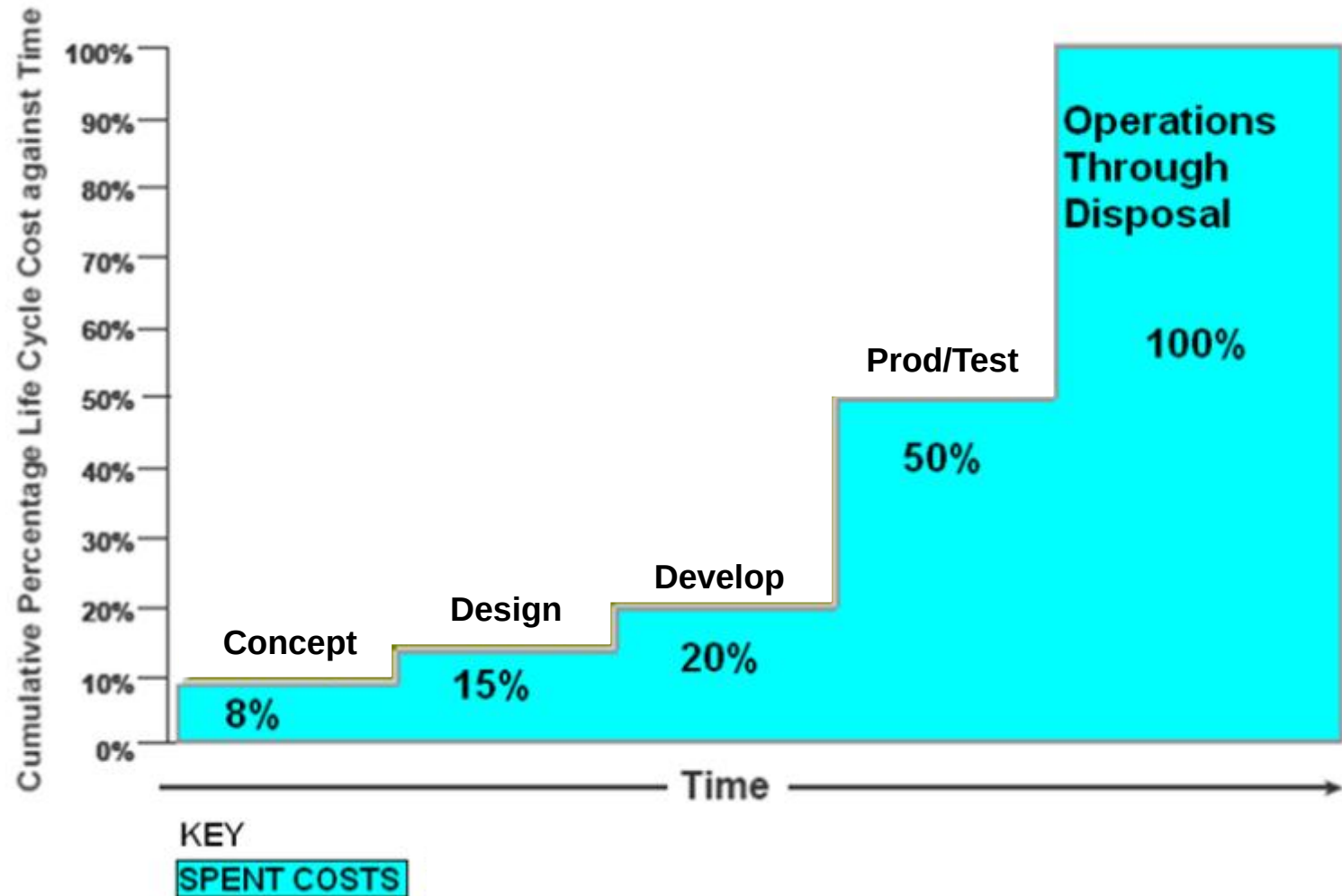
Sources: 1) Leffingwell & Widrig, “Managing Software Requirements,” Addison Wesley, 1999 2) IAG Consulting, 2008 3) Dynamic Market Limited, 2007 4) Standish Group, 2001 5) Don Reinertsen, McKinsey, 1983

Requirements Management Done Well Can...

- Enable reaching market windows sooner
- Enable meeting contractual commitments on time
- Facilitate full compliance with industry standards & regulations
- Facilitate producing products that meet user needs
 - With fewer defects

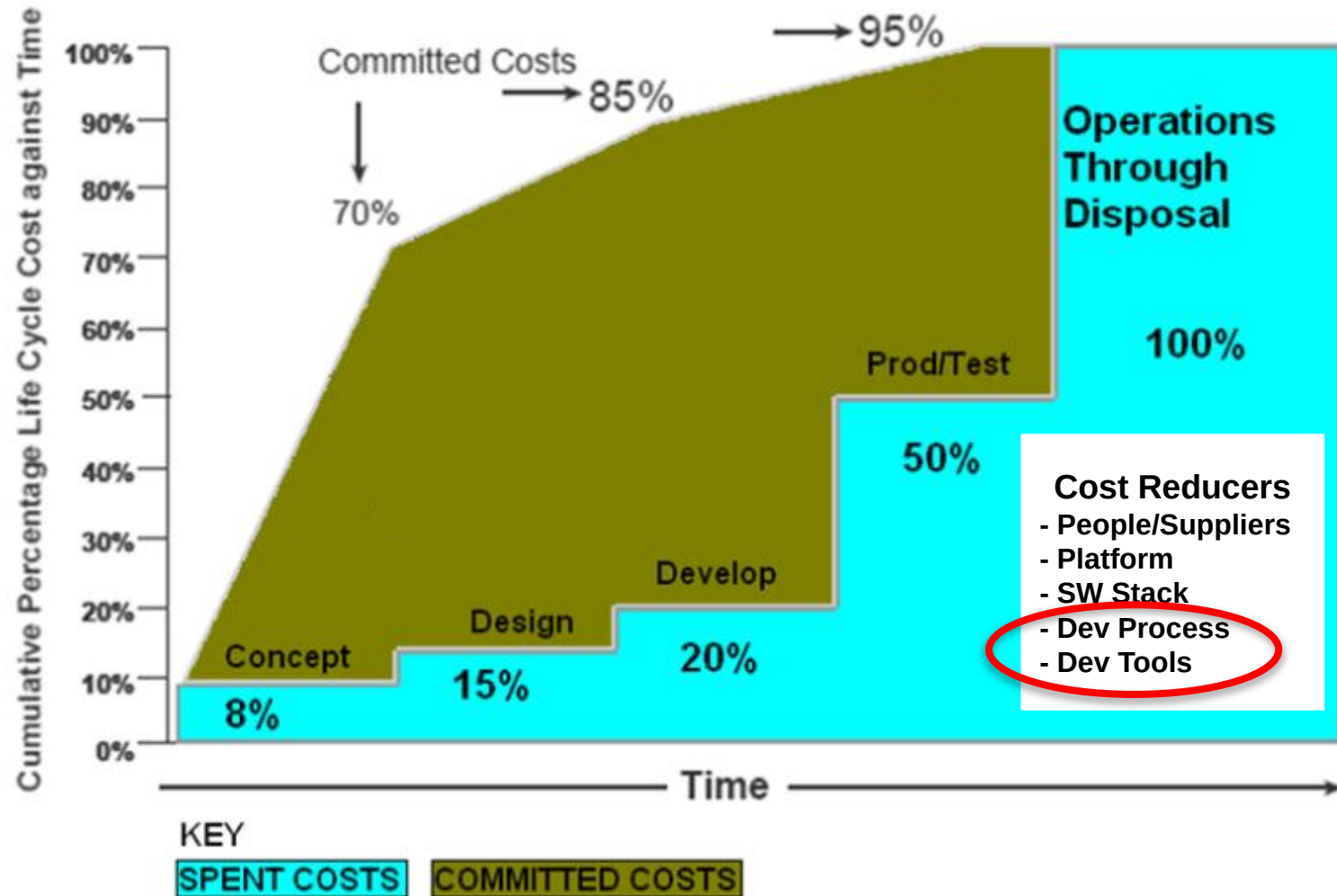
Product Development Lifecycle Influencers

Cumulative Lifecycle Cost – Spent



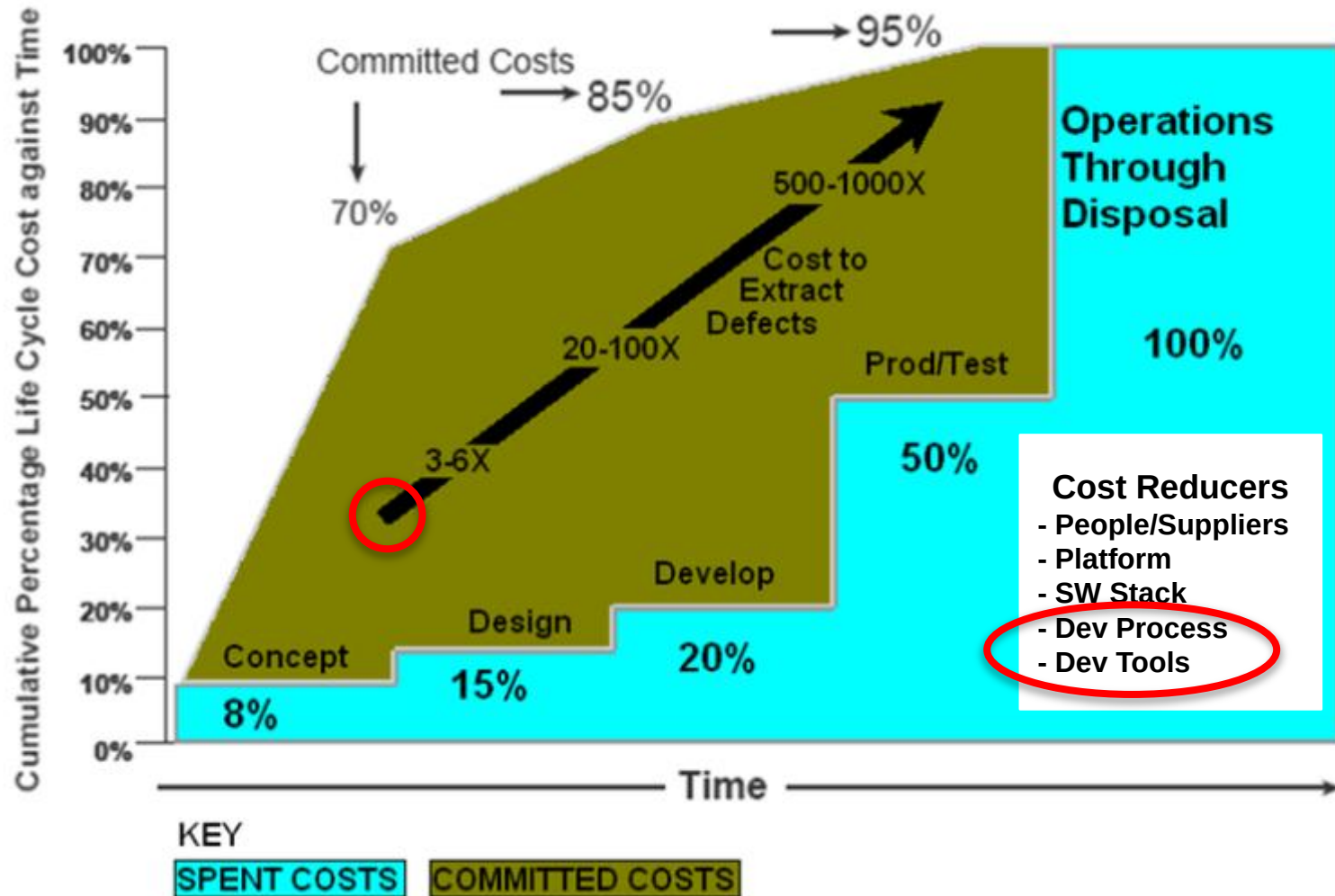
Product Development Lifecycle Influencers

Cumulative Lifecycle Cost – Spent vs. Committed



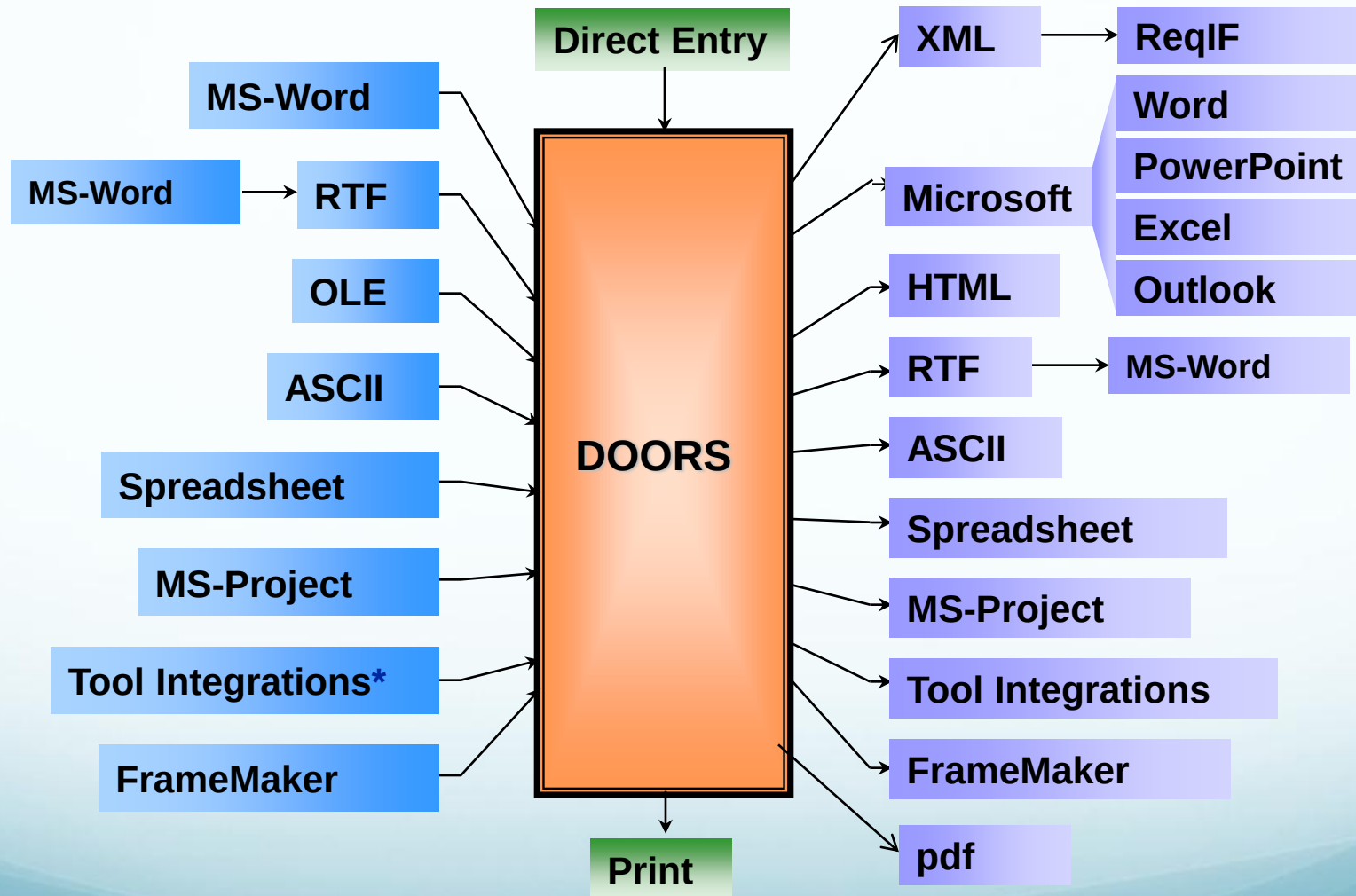
Product Development Lifecycle Influencers

Cumulative Lifecycle Cost – Spent vs. Committed



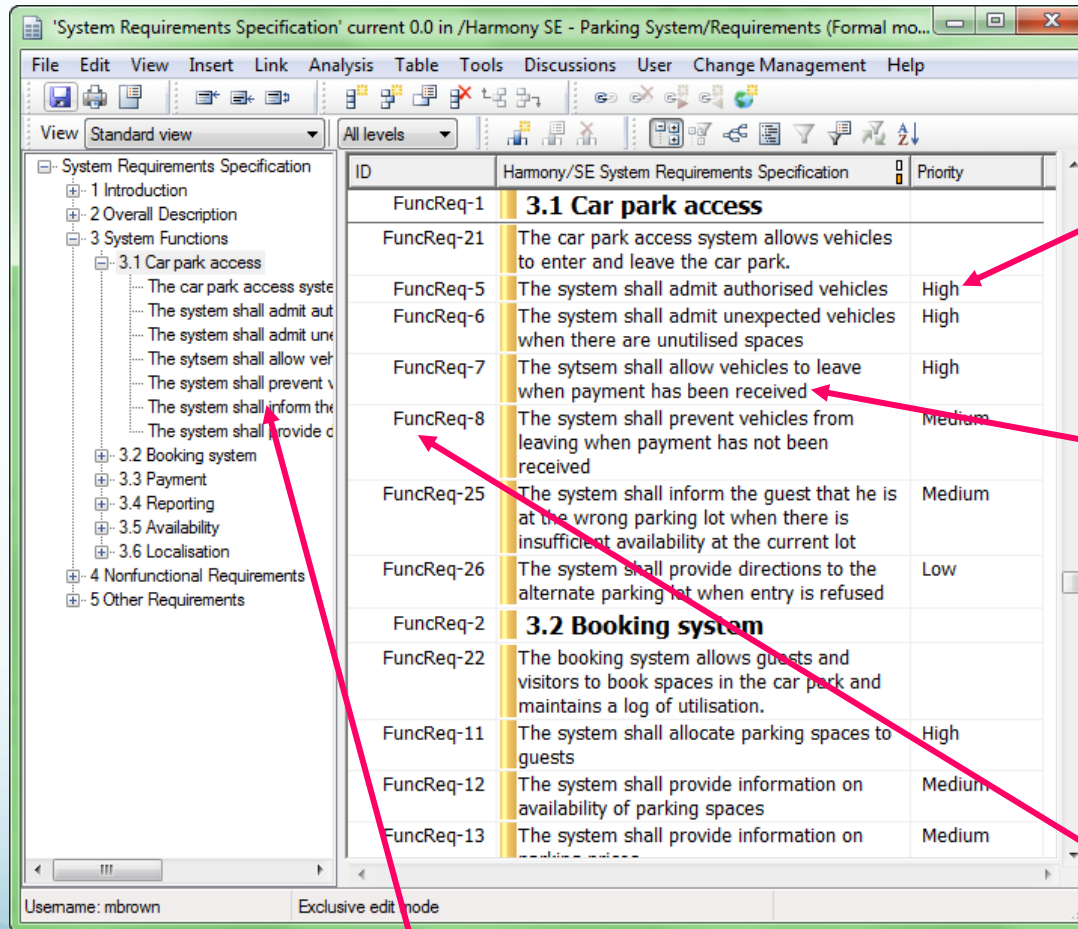
Accelerate Requirements Management

Import All Your Data & Create Documents



Accelerate Requirements Management

DOORS – Like Word and Excel Together



Set attributes like cost, risk, priority accurately

Create traceability to other requirements, tests or information

Reference requirements with precision

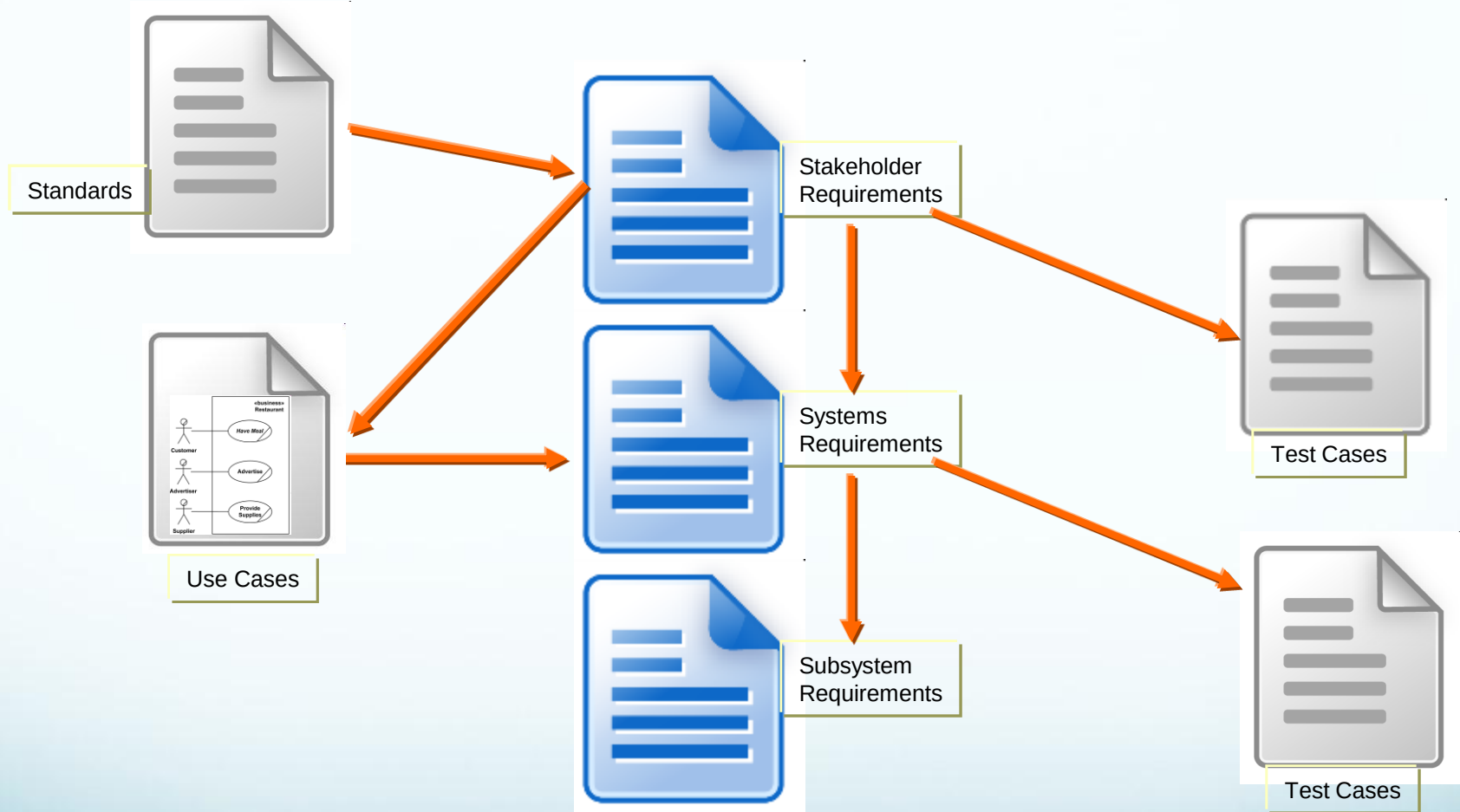
Navigate to specific statements



RIVER NORTH
SOLUTIONS

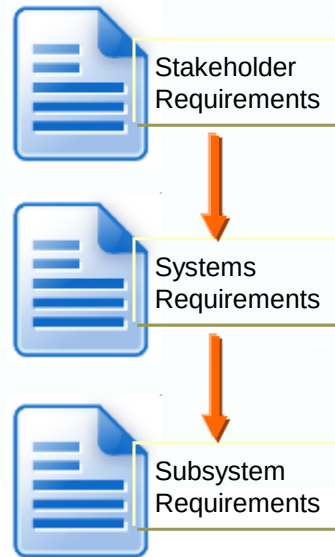
Accelerate Requirements Management

Traceability – Collaboration Across the Lifecycle



Use traceability to manage relationships between linked information

Requirements Management - Traceability



Linked
Stakeholder
Requirements

System
Requirements

Linked
Subsystem
Requirements

	ID	Contents	Satisfied By
		demand from any meter interface via the network or a handheld.	
 923: The meter interface unit shall be able to collect meter data at regular interval (AMR Stakeholder Requirements Specification) Created On: Oct 2, 2012 7:31 PM	929	When connected to a fixed network, the meter interface unit shall 'wake up' and communicate for 4 seconds every 30 minutes, synchronizing all clocks and configuration information. Between these transmissions, the unit remains in a low power state, conserving battery life.	
 844: The meter interface shall detect water leaks and record leak status with the account (AMR Stakeholder Requirements Specification) Created On: Oct 2, 2012 7:31 PM Priority: Medium Schedule: Version 2	871	The meter interface shall detect water leaks and record leak status with the account data.	 805: monitorPressure (Meter Interface Subsystem Requirements Specification) Created On: Oct 2, 2012 7:30 PM Creator: Morgan  806: detectLeak (Meter Interface Subsystem Requirements Specification) Created On: Oct 2, 2012 7:30 PM Creator: Morgan

Accelerate Requirements Management

Attributes – to support and control process

'System Requirements Specification' current 0.0 in /Harmony SE - Parking System/Requirements (Formal module) - DOORS

File Edit View Insert Link Analysis Table Tools Discussions User Change Management Help

View Requirements Default All levels

ID	Harmony/SE System Requirements Specification	Information Type(Rqmt)	Status	Priority	Proposed Verification M
FuncReq-95	3 System Functions	Heading	Unconfirmed		
FuncReq-1	3.1 Car park access	Heading	Not Applicable		
FuncReq-21	The car park access system allows vehicles to enter and leave the car park.	Context	Not Applicable		
FuncReq-5	The system shall admit authorised vehicles	Requirement	Agreed	High	Test
FuncReq-6	The system shall admit unexpected vehicles when there are unutilised spaces	Requirement	Agreed	High	Simulation
FuncReq-7	The system shall allow vehicles to leave when payment has been received	Requirement	Agreed	High	Simulation
FuncReq-8	The system shall prevent vehicles from leaving when payment has not been received	Requirement	Proposed	Medium	Test
FuncReq-25	The system shall inform the guest that he is at the wrong parking lot when there is insufficient availability at the current lot	Requirement	Supp_Assess	Medium	Demonstration
FuncReq-26	The system shall provide directions to the alternate parking lot when entry is refused	Requirement	Stake_Assess	Low	Demonstration
FuncReq-2	3.2 Booking system	Heading	Not Applicable		
FuncReq-22	The booking system allows guests and visitors to book spaces in the car park and maintains a log of utilisation.	Context	Not Applicable		
FuncReq-11	The system shall allocate parking spaces to guests	Requirement	Agreed	High	Test
FuncReq-12	The system shall provide information on availability of parking spaces	Requirement	Unconfirmed	Medium	Test
FuncReq-13	The system shall provide information on parking prices	Requirement	Proposed	Medium	Test
FuncReq-18	The system shall alert registered visitors to the availability of parking spaces	Requirement	Proposed	Low	Test
FuncReq-19	The system shall maintain a number of spaces as unavailable for visitors	Requirement	Proposed	Medium	Simulation
FuncReq-20	The system shall allow the hotel to set the number of reserved spaces for each day	Requirement	Proposed	Medium	Simulation
FuncReq-24	The system shall dynamically reallocate spaces when a guest arrives at the wrong parking lot	Requirement	Agreed	Low	Simulation
FuncReq-3	3.3 Payment	Heading	Not Applicable		
FuncReq-23	The payment system allows guests and visitors to pay for parking by a variety of methods	Context	Not Applicable		
FuncReq-14	The system shall allow guests to pay as part of their hotel check out	Requirement	Agreed	High	Simulation
FuncReq-15	The system shall allow visitors to pay for parking	Requirement	Proposed	Low	Simulation

Username: mbrown Exclusive edit mode

Accelerate Requirements Management

Requirements Management Tool Features

Document structure

Car user requirements
2 User types
2.1 Nationalities
The car will be used in the following countries: UK, USA, Northern Europe, Eastern Europe, Japan, Russia, Australia.
2.2 User sizes
People come in all shapes and sizes. The car must be suitable for people maximum and minimum sizes 1.3 m to 2 m weighing 25 kilograms to 140 kilograms.
3 Requirements
3.1 Capability Requirements
3.1.1 Carrying Capacity
3.1.1.1 Number of people
Four average size adults shall be able to travel in comfort for a period of 4 hours. This level of comfort is defined as being equivalent to the standard of comfort provided by the top 30% of cars produced in 2006.
The top level of cars are those in the price range £13,000 to £30,000 at 2006 prices.
Five average size adults shall be able to travel in comfort for a period of 4 hours.

Attributes

der Requirements' current 1.0 (Issue 1) in /Training Database (Formal module) -		
View Insert Link Analysis Table Tools User Help		
Project progress	All levels	
Car user requirements	Percentage cost	Progress
1 Introduction	0.172835	0
This module contains the user requirements for a new car to be commercially available by 1 August 2006.		0
2 User types	1.370889	0
2.1 Nationalities	0.642687	0
The car will be used in the following	0.769025	0

Filter to focus

Stakeholder Requirements	ID	Car user requirements
1 Introduction	TRN-CSR-3	2 User types
2 User types	TRN-CSR-4	2.1 Nationalities
3 Requirements	TRN-CSR-5	The car will be used in the following countries: UK, USA, Northern Europe, East Japan, Russia, Australia.
3.1 Capability Requirements	TRN-CSR-6	2.2 User sizes
3.1.1 Carrying Capacity	TRN-CSR-7	People come in all shapes and sizes. The car must be suitable for people maximum sizes 1.3 m to 2 m weighing 25 kilograms to 140 kilograms.
3.1.1.1 Number of people		
3.1.1.2 Amount of people		
3.1.2 Cost Points		
3.1.3 Movement		
3.1.4 Fuel economy		
3.1.5 Safety		
3.1.6 Noise levels		
3.1.7 Ease of Access		

View related information

Car user requirements	In-links (System Requirements)
3.1.2.1.1 Forwards	
Users shall be able to travel at speeds up to 200 kilometers per hour.	TRN-SR-5 The car shall be able to move forwards at all speeds from 0 to 220 kilometers per hour on standard flat roads with winds of 0 kilometers per hour, with 280 BHP. Not Set TRN-SR-26 The car shall have a mechanism to enable it to be moved forwards or backwards. Not Set TRN-SR-5

View historical information

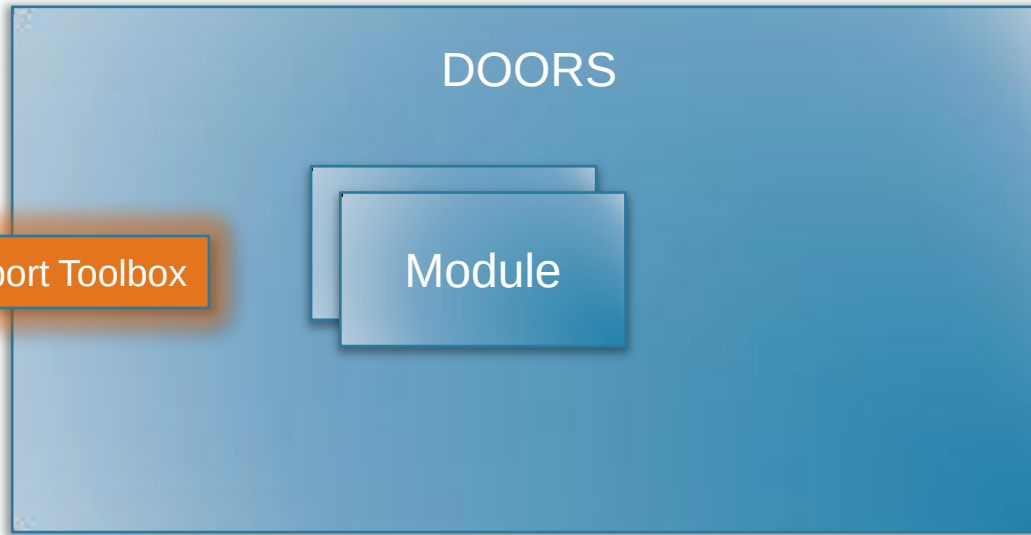
3.1.1.1 Number of people	3.1.1.1 Number of people (Next object differs.) Deleted object 'TRN-CSR-12' follows here:- Four average size adults shall be able to travel in comfort for a period of 4 hours. This level of comfort is defined as being equivalent to the standard of comfort provided by the top 30% of cars produced in 2006.
Five average size adults shall be able to travel in comfort for a period of 4 hours.	Five average size adults shall be able to travel in comfort for a period of 4 hours. (Previous object differs.)
Two average size adults and 3 average size children shall be able to travel in comfort for a period of 3 hours. This could be accomplished with a three seat arrangement.	Two average size adults and 3 average size children shall be able to travel in comfort for a period of 3 hours. This could be accomplished with a three seat arrangement.
This level of comfort required is defined as being equivalent to the standard of comfort provided by the top 30% of cars produced in 2009.	<u>This level of comfort required is defined as being equivalent to the standard of comfort provided by the top 30% of cars produced in 2009.</u>
Users shall have easy entry and exit.	Users shall have easy entry and exit.

Accelerate Requirements Management Workflow View

Word, PDF



Import Toolbox

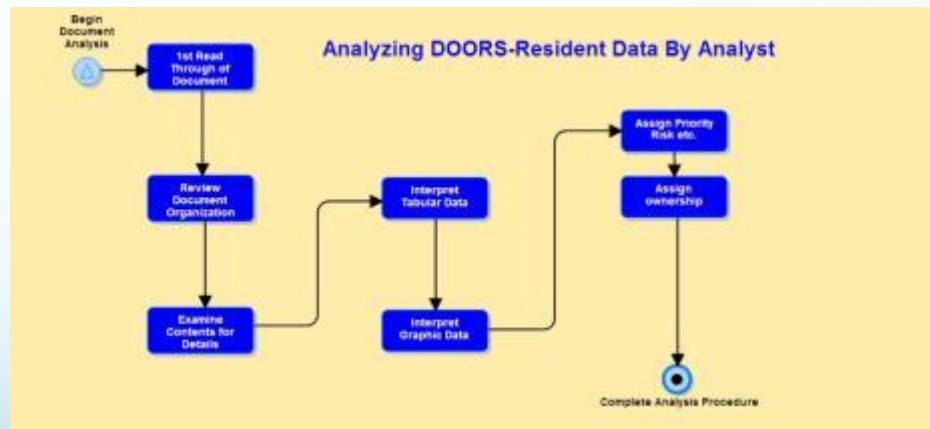
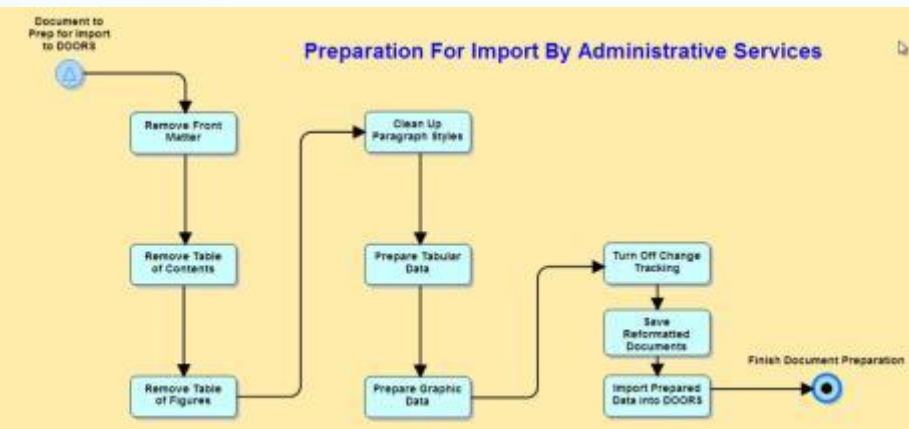
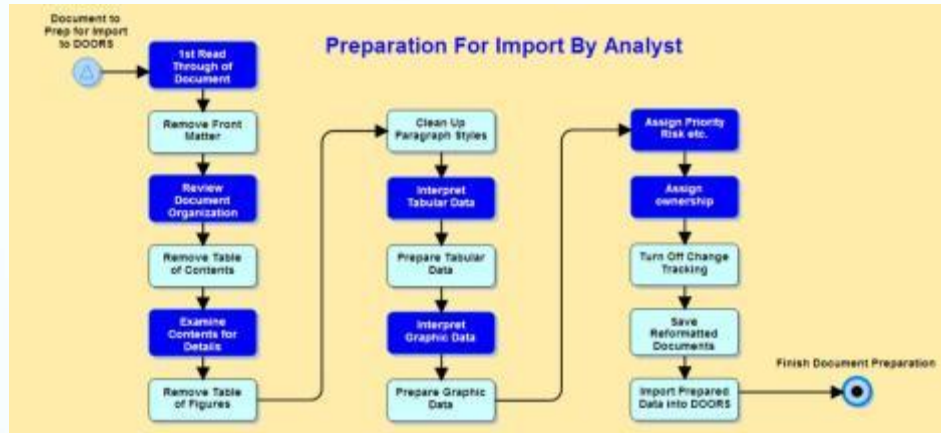


User
imports
data into
DOORS

Accelerate Requirements Management

Preparing Data For DOORS

- Admins excel at formatting
- Uniform DOORS content
- Focus on analysis not form
- Increase productivity



Accelerate Requirements Management

Who Prepares the Document?

Analyst Preps Document

- 4-5+ Days/Document
- Results vary by author
- Impact other tasks
- Impacted by other tasks
- Addresses more issues

Admin Preps Document

- 1-2 Days/Doc Prep
- **Uniform Results**
- Focus on **this** task
- Analyst focuses on imported **results**
- 1-2 days Analysis

Savings can be 1+ days per document

Approach Enables Focus on Data Analysis Not Format



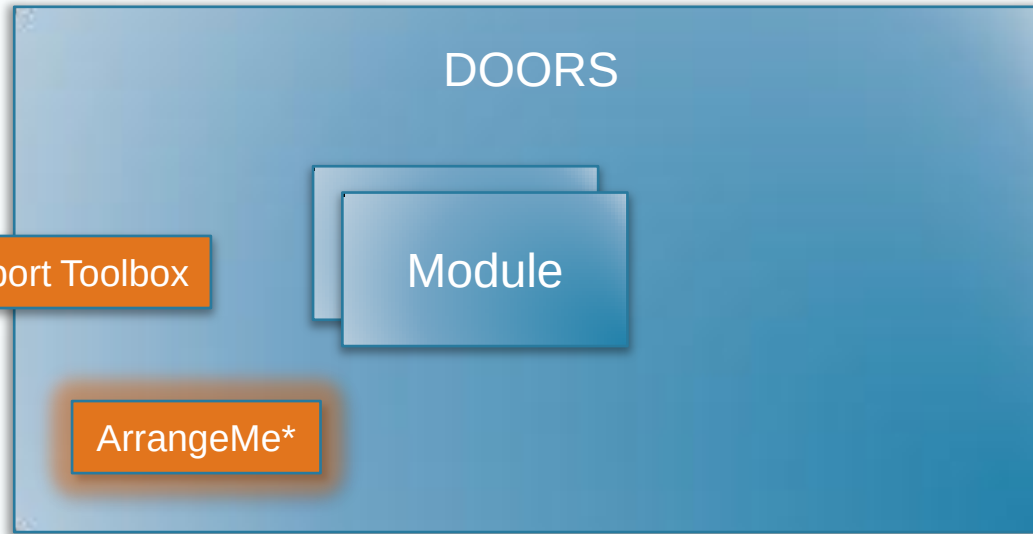
- Engineers, Business Analysts, Managers focus on content
- Apply prioritization, ownership, risk, estimates
- Do not waste time on data cleanup, formatting
- Improves productivity, accelerates projects

Accelerate Requirements Management Workflow View

Word, PDF



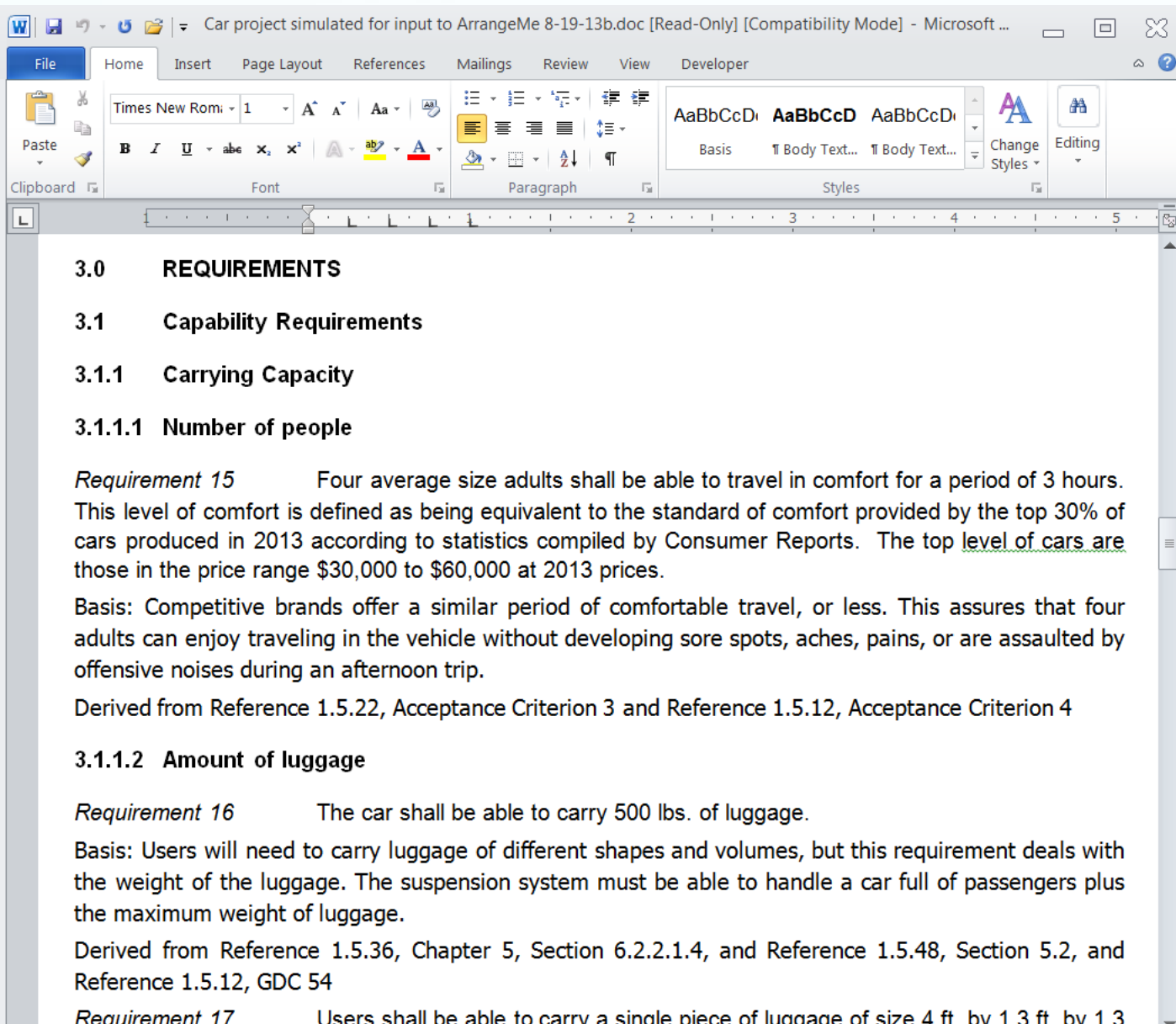
Import Toolbox



*In Beta release

Accelerate Requirements Management

Example of Source Data in Word



3.0 REQUIREMENTS

3.1 Capability Requirements

3.1.1 Carrying Capacity

3.1.1.1 Number of people

Requirement 15 Four average size adults shall be able to travel in comfort for a period of 3 hours. This level of comfort is defined as being equivalent to the standard of comfort provided by the top 30% of cars produced in 2013 according to statistics compiled by Consumer Reports. The top level of cars are those in the price range \$30,000 to \$60,000 at 2013 prices.

Basis: Competitive brands offer a similar period of comfortable travel, or less. This assures that four adults can enjoy traveling in the vehicle without developing sore spots, aches, pains, or are assaulted by offensive noises during an afternoon trip.

Derived from Reference 1.5.22, Acceptance Criterion 3 and Reference 1.5.12, Acceptance Criterion 4

3.1.1.2 Amount of luggage

Requirement 16 The car shall be able to carry 500 lbs. of luggage.

Basis: Users will need to carry luggage of different shapes and volumes, but this requirement deals with the weight of the luggage. The suspension system must be able to handle a car full of passengers plus the maximum weight of luggage.

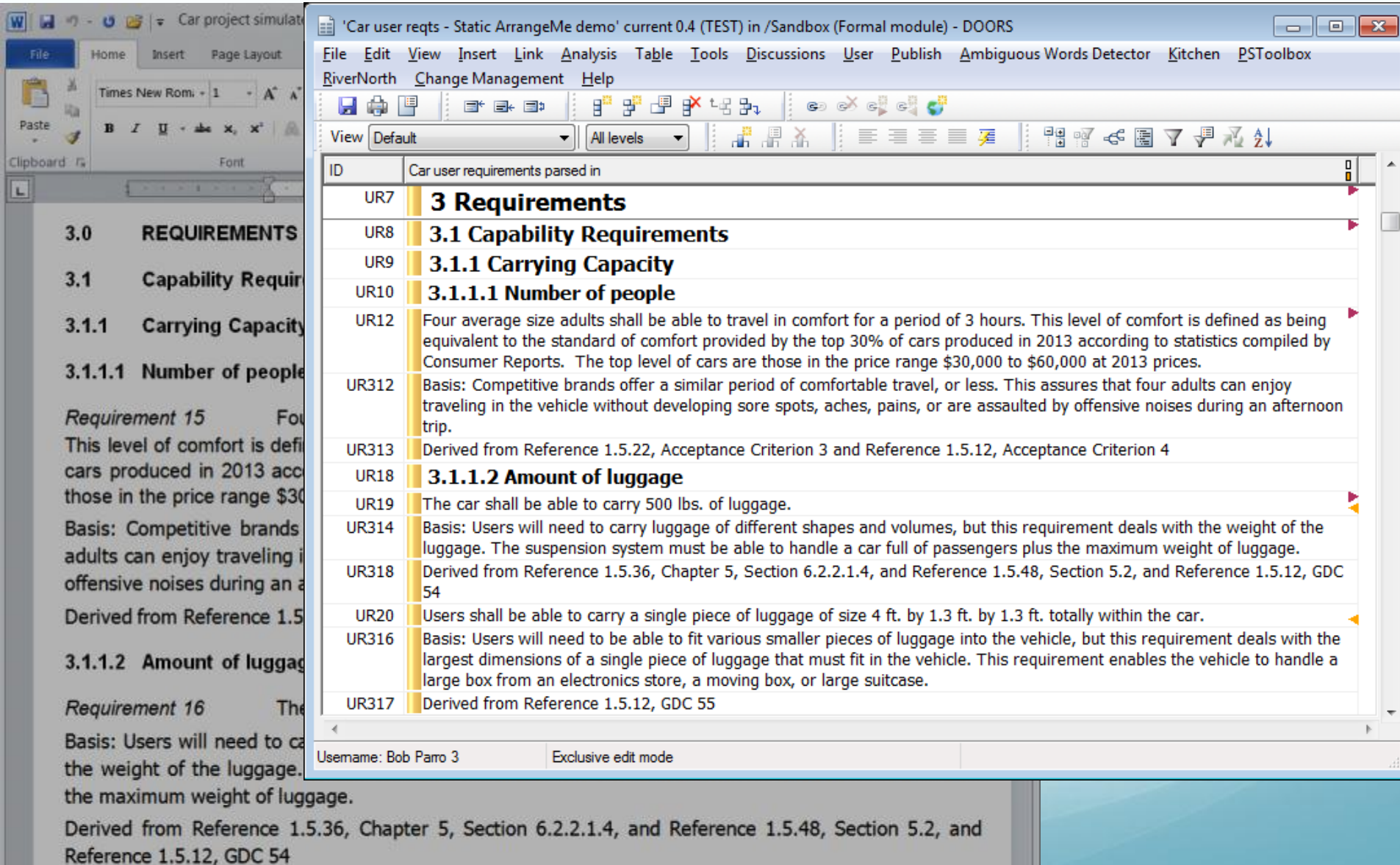
Derived from Reference 1.5.36, Chapter 5, Section 6.2.2.1.4, and Reference 1.5.48, Section 5.2, and Reference 1.5.12, GDC 54

Requirement 17 Users shall be able to carry a single piece of luggage of size 4 ft. by 13 ft. by 13

- Before transferring to DOORS

Accelerate Requirements Management

Word Data Transferred into DOORS



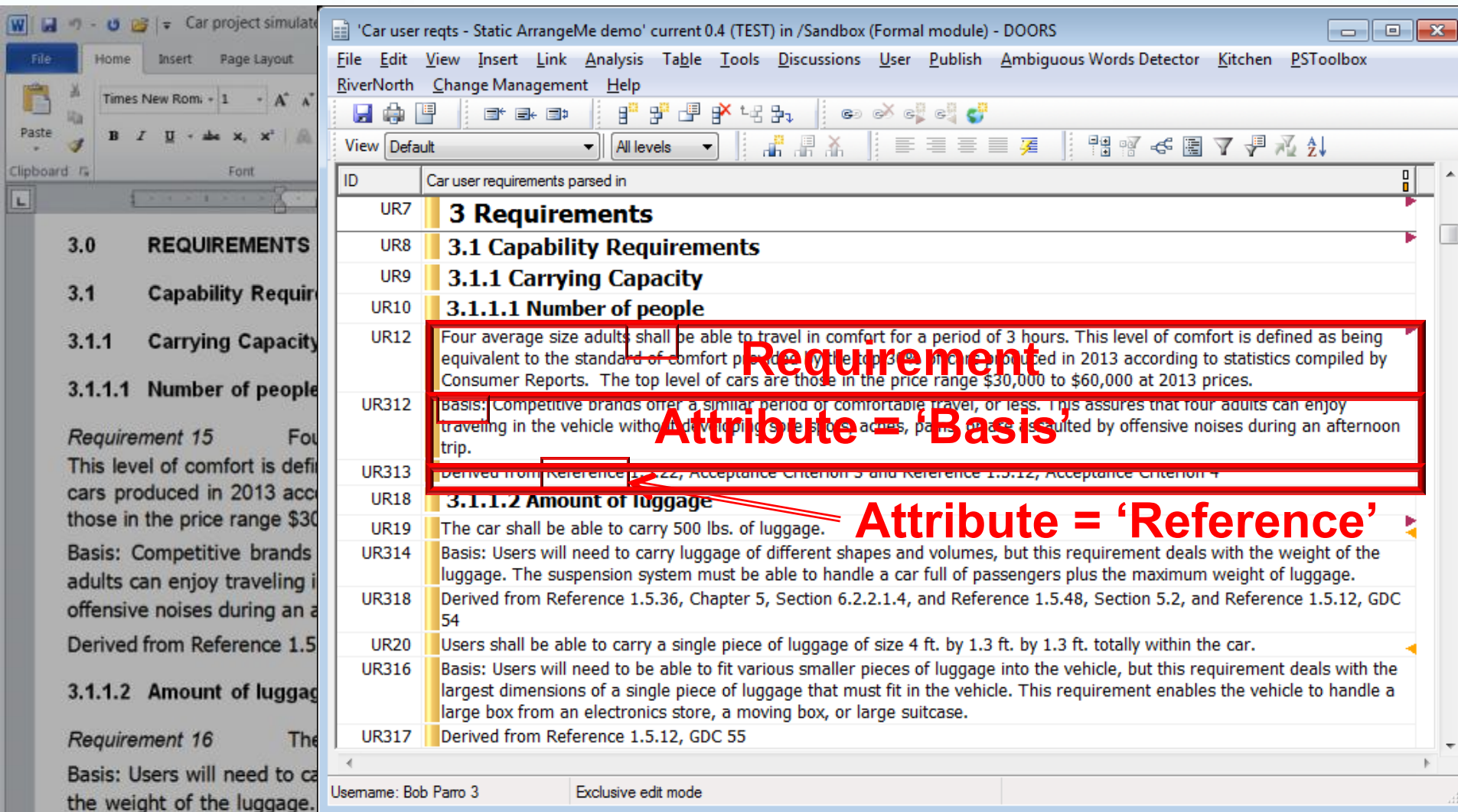
The screenshot shows the DOORS software interface with a list of requirements. The requirements are organized into a hierarchical structure, with each requirement having a unique ID and a description. The requirements are as follows:

ID	Requirement Description
UR7	3 Requirements
UR8	3.1 Capability Requirements
UR9	3.1.1 Carrying Capacity
UR10	3.1.1.1 Number of people
UR12	Four average size adults shall be able to travel in comfort for a period of 3 hours. This level of comfort is defined as being equivalent to the standard of comfort provided by the top 30% of cars produced in 2013 according to statistics compiled by Consumer Reports. The top level of cars are those in the price range \$30,000 to \$60,000 at 2013 prices.
UR312	Basis: Competitive brands offer a similar period of comfortable travel, or less. This assures that four adults can enjoy traveling in the vehicle without developing sore spots, aches, pains, or are assaulted by offensive noises during an afternoon trip.
UR313	Derived from Reference 1.5.22, Acceptance Criterion 3 and Reference 1.5.12, Acceptance Criterion 4
UR18	3.1.1.2 Amount of luggage
UR19	The car shall be able to carry 500 lbs. of luggage.
UR314	Basis: Users will need to carry luggage of different shapes and volumes, but this requirement deals with the weight of the luggage. The suspension system must be able to handle a car full of passengers plus the maximum weight of luggage.
UR318	Derived from Reference 1.5.36, Chapter 5, Section 6.2.2.1.4, and Reference 1.5.48, Section 5.2, and Reference 1.5.12, GDC 54
UR20	Users shall be able to carry a single piece of luggage of size 4 ft. by 1.3 ft. by 1.3 ft. totally within the car.
UR316	Basis: Users will need to be able to fit various smaller pieces of luggage into the vehicle, but this requirement deals with the largest dimensions of a single piece of luggage that must fit in the vehicle. This requirement enables the vehicle to handle a large box from an electronics store, a moving box, or large suitcase.
UR317	Derived from Reference 1.5.12, GDC 55

The software interface also shows a menu bar with options like File, Edit, View, Insert, Link, Analysis, Table, Tools, Discussions, User, Publish, Ambiguous Words Detector, Kitchen, and PSToolbox. The status bar at the bottom indicates the user is 'Bob Parro 3' and the software is in 'Exclusive edit mode'.

Accelerate Requirements Management

Use Patterns to Identify Data to be Manipulated



The screenshot shows the DOORS software interface. The left pane displays a requirements tree with the following structure:

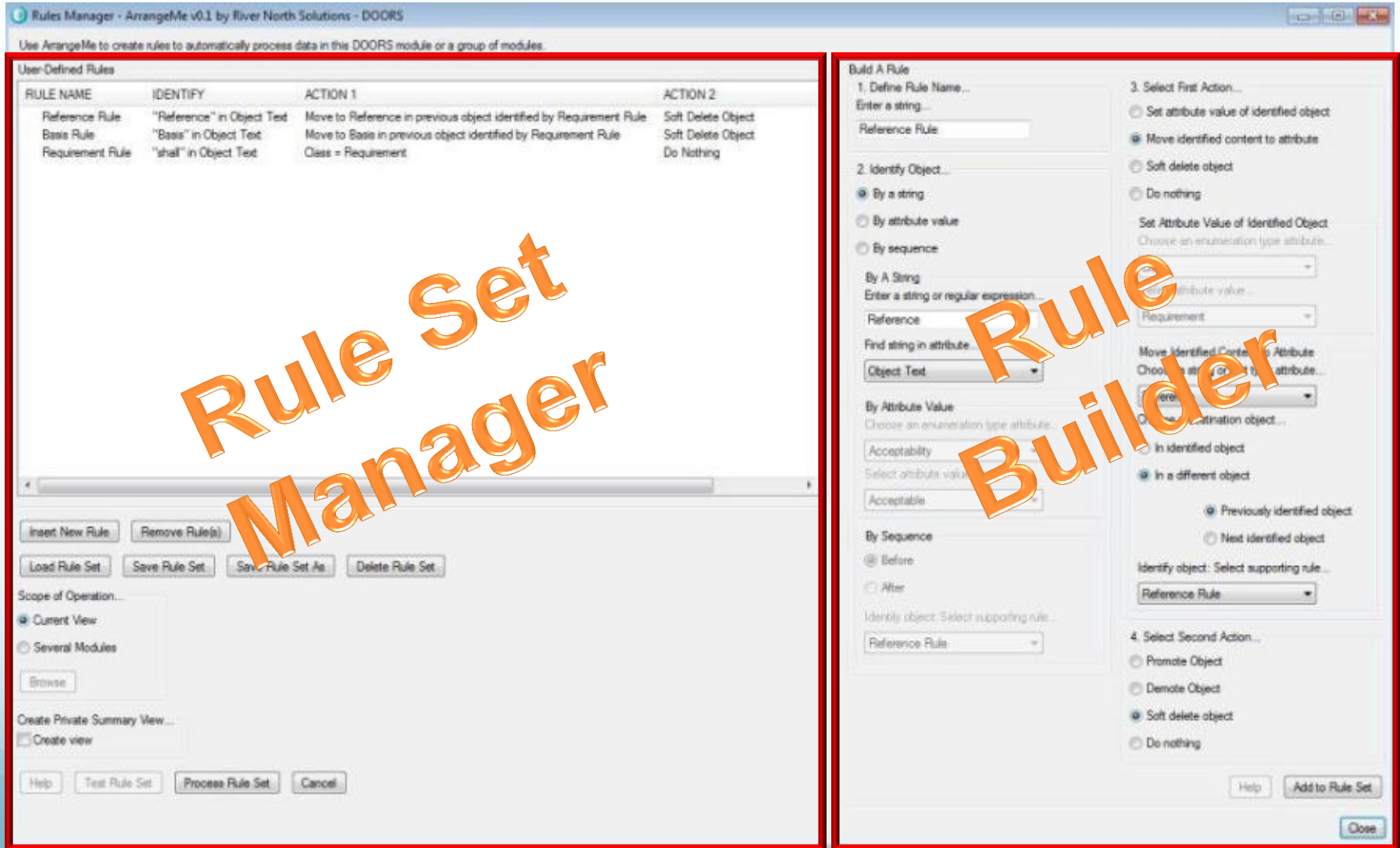
- 3.0 REQUIREMENTS
 - 3.1 Capability Requirements
 - 3.1.1 Carrying Capacity
 - 3.1.1.1 Number of people

The right pane shows a list of requirements. The following requirements are highlighted with red boxes and annotations:

- UR7: 3 Requirements
- UR8: 3.1 Capability Requirements
- UR9: 3.1.1 Carrying Capacity
- UR10: 3.1.1.1 Number of people
- UR12: Four average size adults shall be able to travel in comfort for a period of 3 hours. This level of comfort is defined as being equivalent to the standard of comfort provided by the top 30% of cars produced in 2013 according to statistics compiled by Consumer Reports. The top level of cars are those in the price range \$30,000 to \$60,000 at 2013 prices.
 - Annotation: Requirement
- UR312: Basis: Competitive brands offer a similar period of comfortable travel, or less. This assures that four adults can enjoy traveling in the vehicle without discomfort.
 - Annotation: Attribute = 'Basis'
- UR313: Derived from Reference 1.5.22, Acceptance Criterion 3 and Reference 1.5.12, Acceptance Criterion 4
- UR18: 3.1.1.2 Amount of luggage
- UR19: The car shall be able to carry 500 lbs. of luggage.
 - Annotation: Attribute = 'Reference'
- UR314: Basis: Users will need to carry luggage of different shapes and volumes, but this requirement deals with the weight of the luggage. The suspension system must be able to handle a car full of passengers plus the maximum weight of luggage.
- UR318: Derived from Reference 1.5.36, Chapter 5, Section 6.2.2.1.4, and Reference 1.5.48, Section 5.2, and Reference 1.5.12, GDC 54
- UR20: Users shall be able to carry a single piece of luggage of size 4 ft. by 1.3 ft. by 1.3 ft. totally within the car.
- UR316: Basis: Users will need to be able to fit various smaller pieces of luggage into the vehicle, but this requirement deals with the largest dimensions of a single piece of luggage that must fit in the vehicle. This requirement enables the vehicle to handle a large box from an electronics store, a moving box, or large suitcase.
- UR317: Derived from Reference 1.5.12, GDC 55

The bottom status bar shows: Username: Bob Parro 3 Exclusive edit mode

Build Rules to Automate Data Manipulation



Rules Manager - ArrangeMe v0.1 by River North Solutions - DOORS

Use ArrangeMe to create rules to automatically process data in this DOORS module or a group of modules.

User-Defined Rules

RULE NAME	IDENTIFY	ACTION 1	ACTION 2
Reference Rule	"Reference" in Object Text	Move to Reference in previous object identified by Requirement Rule	Soft Delete Object
Basis Rule	"Basis" in Object Text	Move to Basis in previous object identified by Requirement Rule	Soft Delete Object
Requirement Rule	"shall" in Object Text	Class = Requirement	Do Nothing

Insert New Rule Remove Rule(s)

Load Rule Set Save Rule Set Save Rule Set As Delete Rule Set

Scope of Operation...

☒ Current View

☐ Several Modules

Browse

Create Private Summary View...

☐ Create view

Help Test Rule Set Process Rule Set Cancel

Build A Rule

1. Define Rule Name...

Enter a string...

Reference Rule

2. Identify Object...

☒ By a string

☐ By attribute value

☐ By sequence

By A String

Enter a string or regular expression...

Reference

Find string in attribute...

Object Text

By Attribute Value

Choose an enumeration type attribute...

Acceptability

Select attribute value...

Acceptable

By Sequence

☒ Before

☐ After

Identify object: Select supporting rule...

Reference Rule

3. Select First Action...

☐ Set attribute value of identified object

☒ Move identified content to attribute

☐ Soft delete object

☐ Do nothing

Set Attribute Value of Identified Object

Choose an enumeration type attribute...

Requirement

Move Identified Content to Attribute

Choose an attribute...

Requirement

Choose an enumeration object...

☒ In identified object

☐ In a different object

☒ Previously identified object

☐ Next identified object

Identify object: Select supporting rule...

Reference Rule

4. Select Second Action...

☐ Promote Object

☐ Demote Object

☒ Soft delete object

☐ Do nothing

Help Add to Rule Set

Close



RIVER NORTH
SOLUTIONS

ArrangeMe

ArrangeMe Rule Set

Rules Manager - ArrangeMe v0.1 by River North Solutions - DOORS

Use ArrangeMe to create rules to automatically process data in this DOORS module or a group of modules.

Pre-Defined Rules

RULE NAME	IDENTIFY	ACTION 1	ACTION 2
Reference Rule	"Reference" in Object Text	Move to Reference in previous object identified by Requirement Rule	Soft Delete Object
Basis Rule	"Basis" in Object Text	Move to Basis in previous object identified by Requirement Rule	Soft Delete Object
Requirement Rule	"shall" in Object Text	Class = Requirement	Do Nothing

Build A Rule

1. Define Rule Name...
Enter a string...
Reference Rule

2. Identify Object...
☒ By a string
☐ By attribute value

3. Select First Action...
☐ Set attribute value of identified object
☒ Move identified content to attribute
☐ Soft delete object
☐ Do nothing

User-Defined Rules

RULE NAME	IDENTIFY	ACTION 1	ACTION 2
Reference Rule	"Reference" in Object Text	Move to Reference in previous object identified by Requirement Rule	Soft Delete Object
Basis Rule	"Basis" in Object Text	Move to Basis in previous object identified by Requirement Rule	Soft Delete Object
Requirement Rule	"shall" in Object Text	Class = Requirement	Do Nothing

Insert New Rule Remove Rule(s)

Load Rule Set Save Rule Set Save Rule Set As Delete Rule Set

Scope of Operation...
☒ Current View
☐ Several Modules

Create Private Summary View...
☐ Create view

Help Test Rule Set Process Rule Set Cancel

Acceptability
Select attribute value...
Acceptable

By Sequence
☒ Before
☐ After

Identify object: Select supporting rule...
Reference Rule

☐ In identified object
☒ In a different object
☒ Previously identified object
☐ Next identified object

Identify object: Select supporting rule...
Reference Rule

4. Select Second Action...
☐ Promote Object
☐ Demote Object
☒ Soft delete object
☐ Do nothing

Help Add to Rule Set

Close

Beginning State Before Processing a Module

'Car user reqts' current 0.4 (TEST) in /Car Project for Testing (Formal module) - DOORS

File Edit View Insert Link Analysis Table Tools Discussions User Publish Ambiguous Words Detector Kitchen PSToolbox RiverNorth Change Management Help

View Standard view All levels

ID	Car user requirements parsed in	Basis	Reference	Class
UR24	3.1.2 Movement			
UR25	3.1.2.1 Speed and Acceleration			
UR26	3.1.2.1.1 Forwards			
UR27	Users shall be able to travel at speeds up to 200 miles per hour.	▶		
UR316	Basis: Some jurisdictions allow high speeds, such as Germany.			
UR317	Reference: Reference 8, Section (c)(3)			
UR28	Users shall be able to accelerate from 0 to 100 miles per hour in 10 seconds.	▶		
UR318	Basis: Need to compete with acceleration performance of top 30% of luxury sedans on the market.			
UR319	Reference: Reference 8, Section (c)(4)(i)			
UR29	Users shall be able to travel automatically at predefined speeds.	▶		
UR320	Basis: Cruise control is standard on top 70% of sedans on the market.			
UR321	Reference: Reference 7, Section III, D, 1, a and III, D,1, b			
UR31	3.1.2.1.2 Backwards	▶		
UR32	Users shall be able to move backwards to a maximum speed of 20 miles per hour.	▶		
UR322	Basis: Rearward motion is a secondary function of the vehicle and high speeds are not needed.			
UR324	Reference: Reference 7, Section III, D, 2, a			
UR33	3.1.2.2 Distance			
UR34	Users shall be able to travel 1000 miles without the need for any form of additional fuel.			
UR128	Users shall be able to travel 1500 miles without the need for any form of additional fuel.			

Username: Bob Parro 3 Exclusive edit mode

After ArrangeMe Processes Module (using example rules)

'Car user reqts' current 0.4 (TEST) in /Car Project for Testing (Formal module) - DOORS

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View Standard view All levels

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UR316	✗ Basis: Some jurisdictions allow high speeds, such as Germany.			
UR317	✗ Reference: Reference 8, Section (c)(3)			
UR28	Users shall be able to accelerate from 0 to 100 miles per hour in 10 seconds.	▶ Basis: Need to compete with acceleration performance of top 30% of luxury sedans on the market.	Reference: Reference 8, Section (c)(4)(i)	Requirement
UR318	✗ Basis: Need to compete with acceleration performance of top 30% of luxury sedans on the market.			
UR319	✗ Reference: Reference 8, Section (c)(4)(i)			
UR29	Users shall be able to travel automatically at predefined speeds.	▶ Basis: Cruise control is standard on top 70% of sedans on the market.	Reference: Reference 7, Section III, D, 1, a and III, D,1, b	Requirement
UR320	✗ Basis: Cruise control is standard on top 70% of sedans on the market.			
UR321	✗ Reference: Reference 7, Section III, D, 1, a and III, D,1, b			
UR31	3.1.2.1.2 Backwards			
UR32	Users shall be able to move backwards to a maximum speed of 20 miles per hour.	▶ Basis: Rearward motion is a secondary function of the	Reference: Reference 7, Section III, D, 2, a	Requirement

Username: Bob Parno 3 Exclusive edit mode



RIVER NORTH
SOLUTIONS

Accelerate Requirements Management

ArrangeMe™ Rule Set Execution

User-Defined Rules			
RULE NAME	IDENTIFY	ACTION 1	ACTION 2
Reference Rule	"Reference" in Object Text	Move to Reference in previous object identified by Requirement Rule	Soft Delete Object
Basis Rule	"Basis" in Object Text	Move to Basis in previous object identified by Requirement Rule	Soft Delete Object
Requirement Rule	"shall" in Object Text	Class = Requirement	Do Nothing

'Car user reqts' current 0.4 (TEST) in /Car Project for Testing (Formal module) - DOORS				
File Edit View Insert Link Analysis Table Tools Discussions User Publish Ambiguous Words Detector Kitchen PSToolbox RiverNorth Change Management Help				
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UR320	Basis: Cruise control is standard on top 70% of sedans on the market.			
UR321	Reference: Reference 7, Section III, D, 1, a and III, D,1, b			
UR331	3.1.2.1.2 Backwards			

Accelerate Requirements Management

ArrangeMe™ Rule Set Execution

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UR321	3.1.2.1.2 Backwards			

←-Soft deleted, copied to nearest previous rqmt

Accelerate Requirements Management

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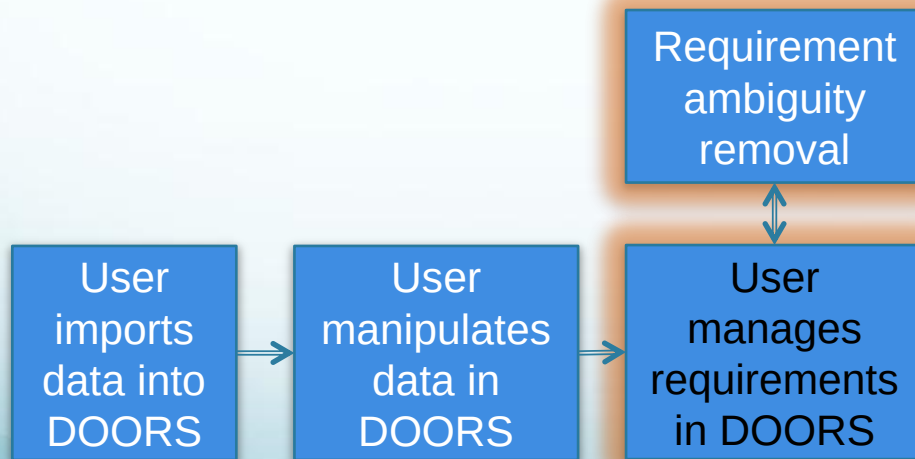
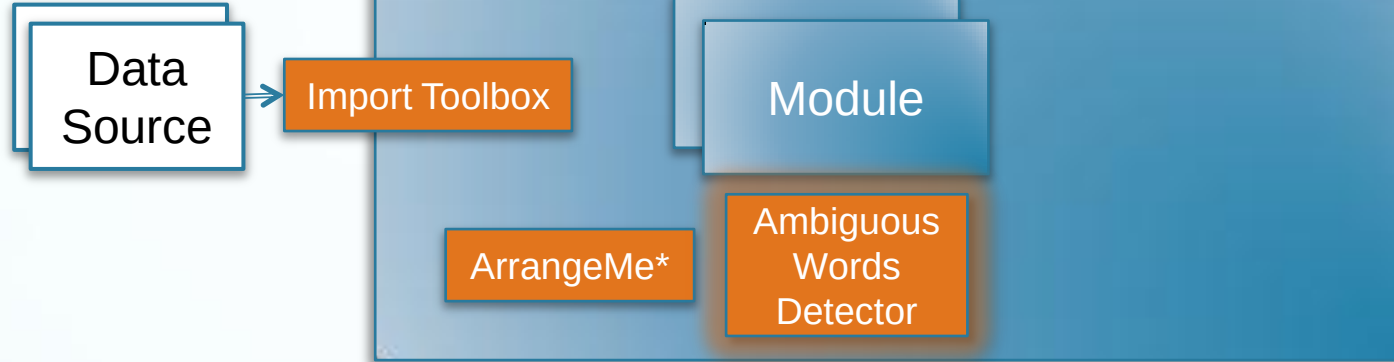
Accelerate Requirements Management

ArrangeMe Value Proposition

- Westinghouse Nuclear
 - Thousands of documents for a power plant project
 - Many hundreds of pages per document
 - Import/data manipulation
 - **3 weeks/document -> 2 hours/document**
- Flexible UI allows users to define data manipulation rules unique to their needs
- Rule sets stored centrally in DOORS for all users to leverage

Accelerate Requirements Management Workflow View

Word, PDF

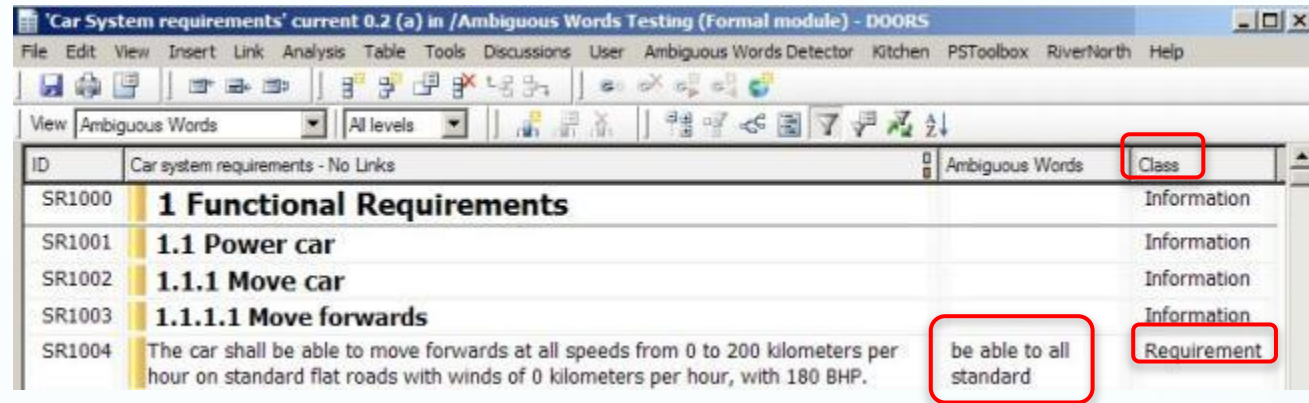


*In Beta release

Accelerate Requirements Management

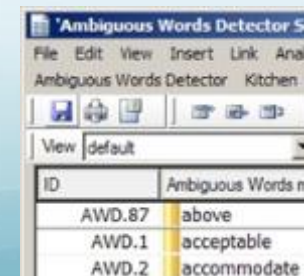
Ambiguous Words Detector (AWD)

- Automatically detects usage of ambiguous words in requirements



ID	Car system requirements - No Links	Ambiguous Words	Class
SR1000	1 Functional Requirements		Information
SR1001	1.1 Power car		Information
SR1002	1.1.1 Move car		Information
SR1003	1.1.1.1 Move forwards		Information
SR1004	The car shall be able to move forwards at all speeds from 0 to 200 kilometers per hour on standard flat roads with winds of 0 kilometers per hour, with 180 BHP.	be able to all standard	Requirement

- Blends into your DOORS schema
 - Recognizes requirements from attribute/enumerated value used in your schema to define object as requirement
 - New:** allows >1 attribute/enum value to define object as rqmt
- Ambiguous words search list contained in DOORS module
 - No coding required to modify list



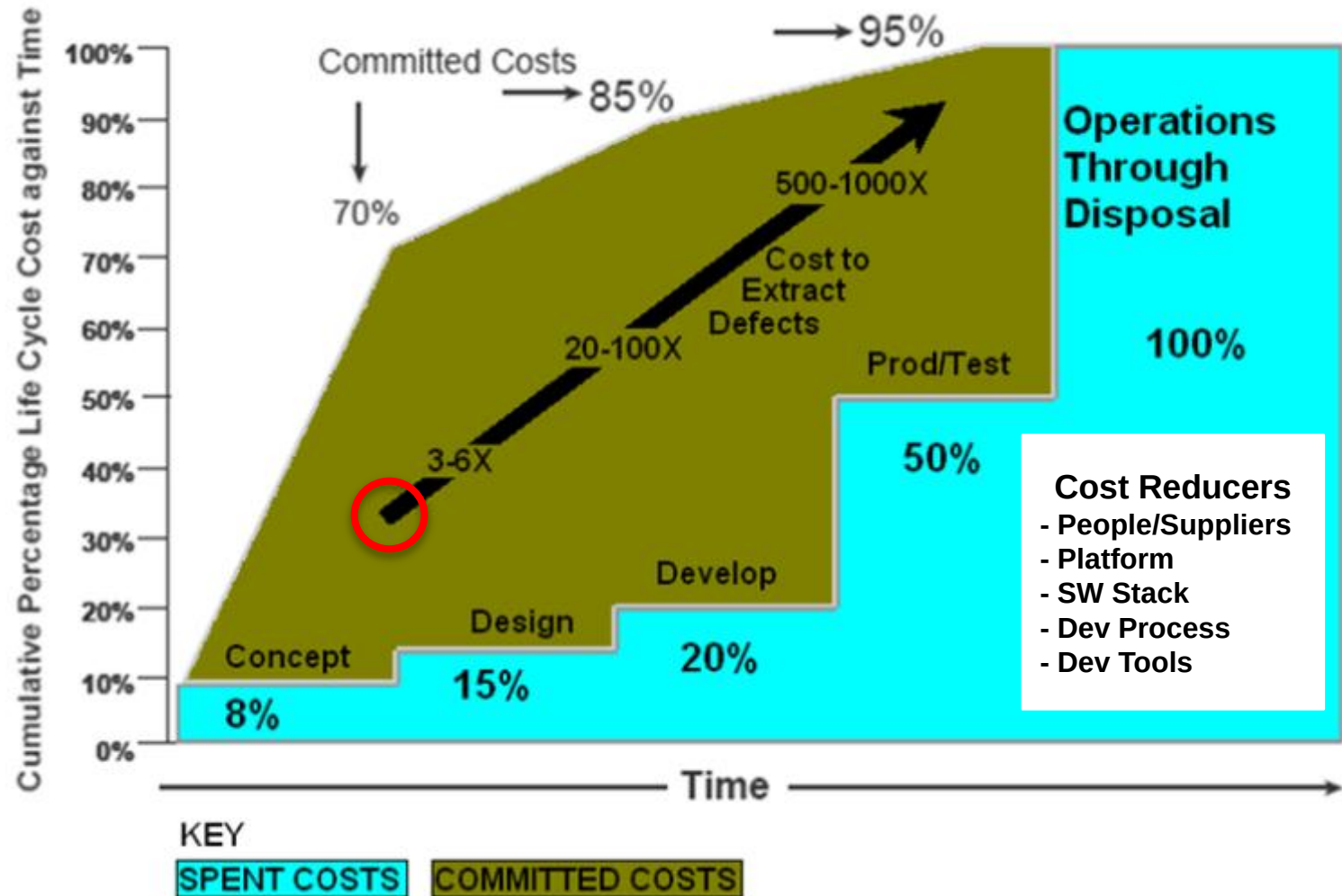
ID	Ambiguous Words m
AWD.87	above
AWD.1	acceptable
AWD.2	accommodate

Accelerate Requirements Management

Value of Eliminating Ambiguity Early

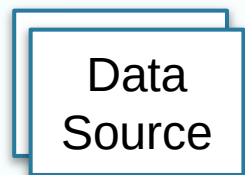
CHURN REDUCTION

Impact of early defect removal

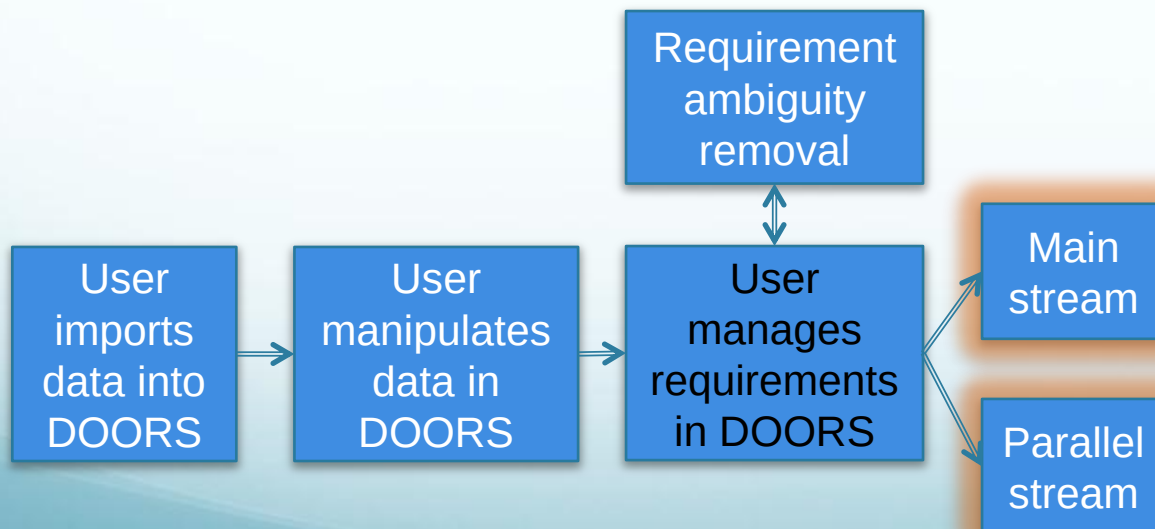
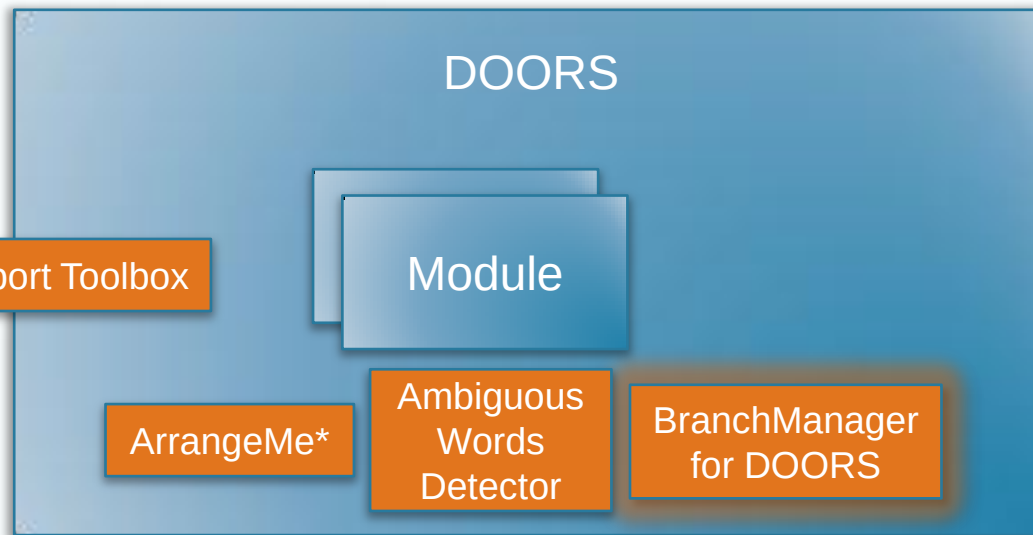


Accelerate Requirements Management Workflow View

Word, PDF



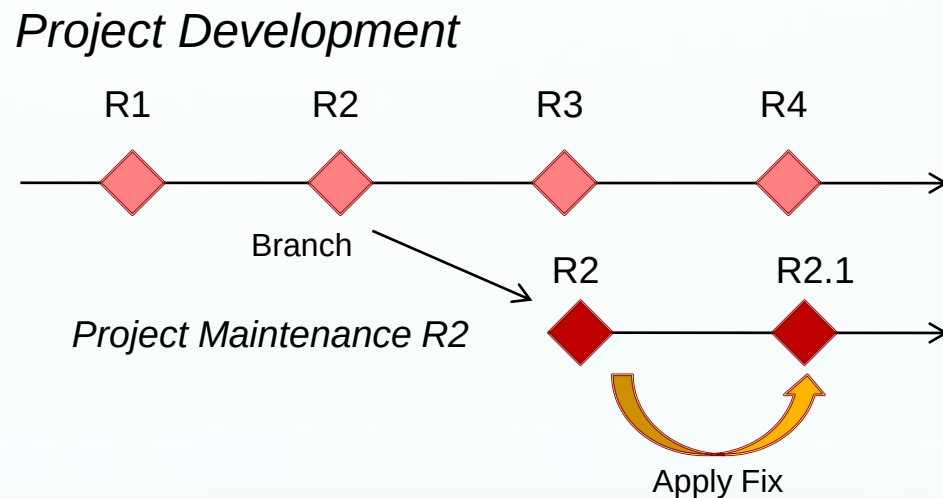
Import Toolbox



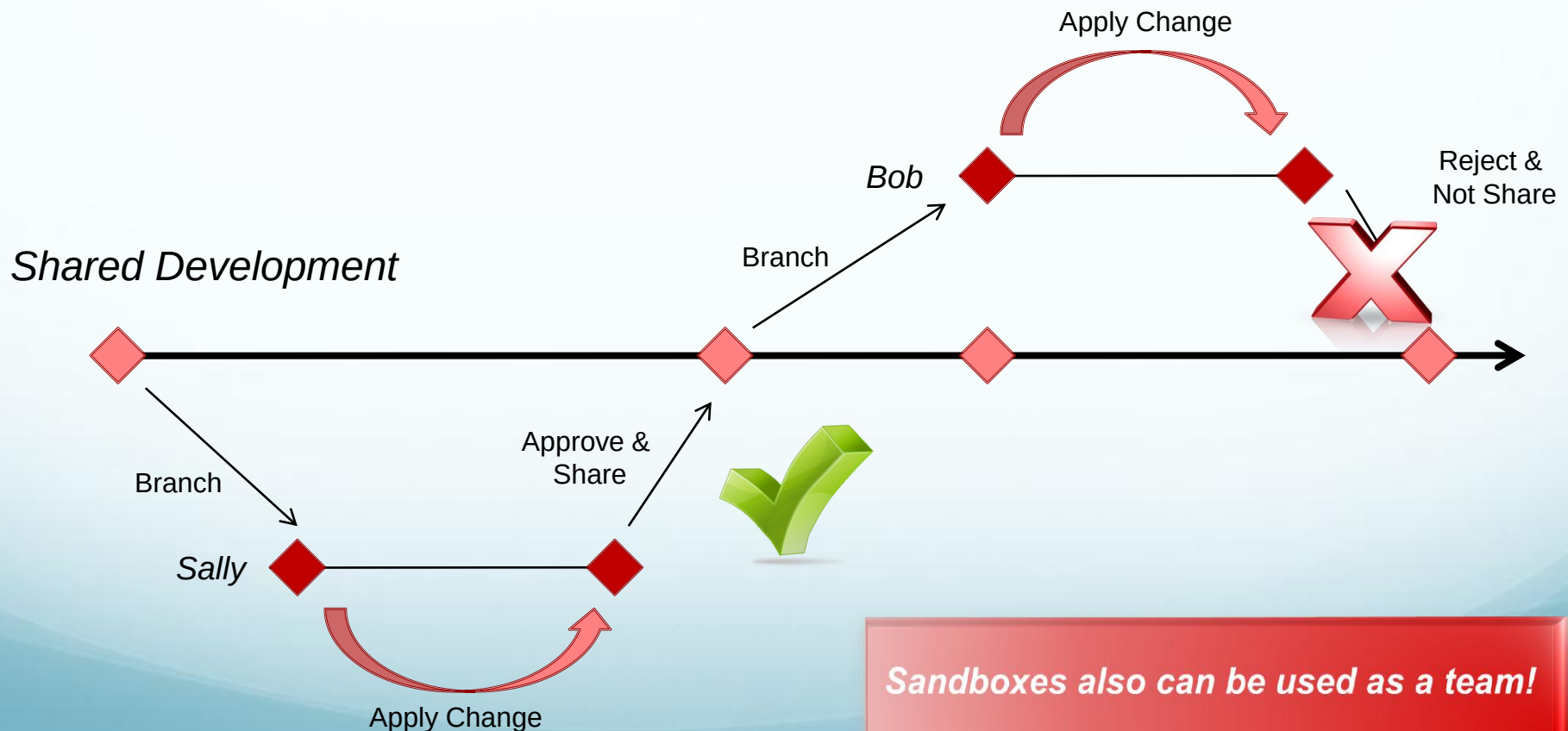
*In Beta release

- Why needed?
 - Derivative product
 - Maintenance branches
 - Fix and propagate
 - Core requirement reuse

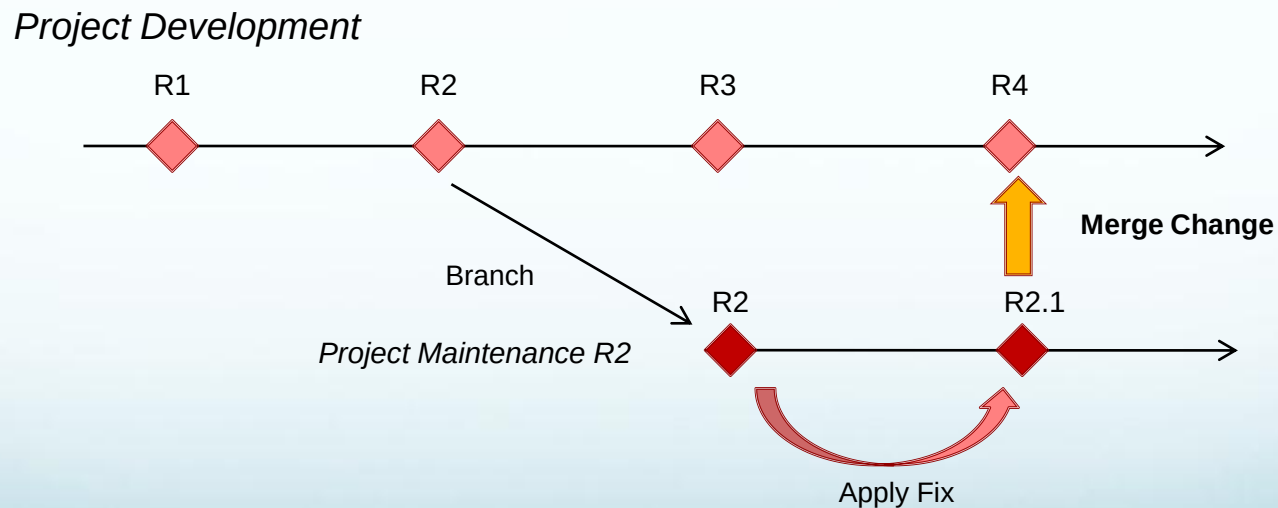
- Maintenance branching



- Maintenance branching
- Personal sandbox & change evaluation



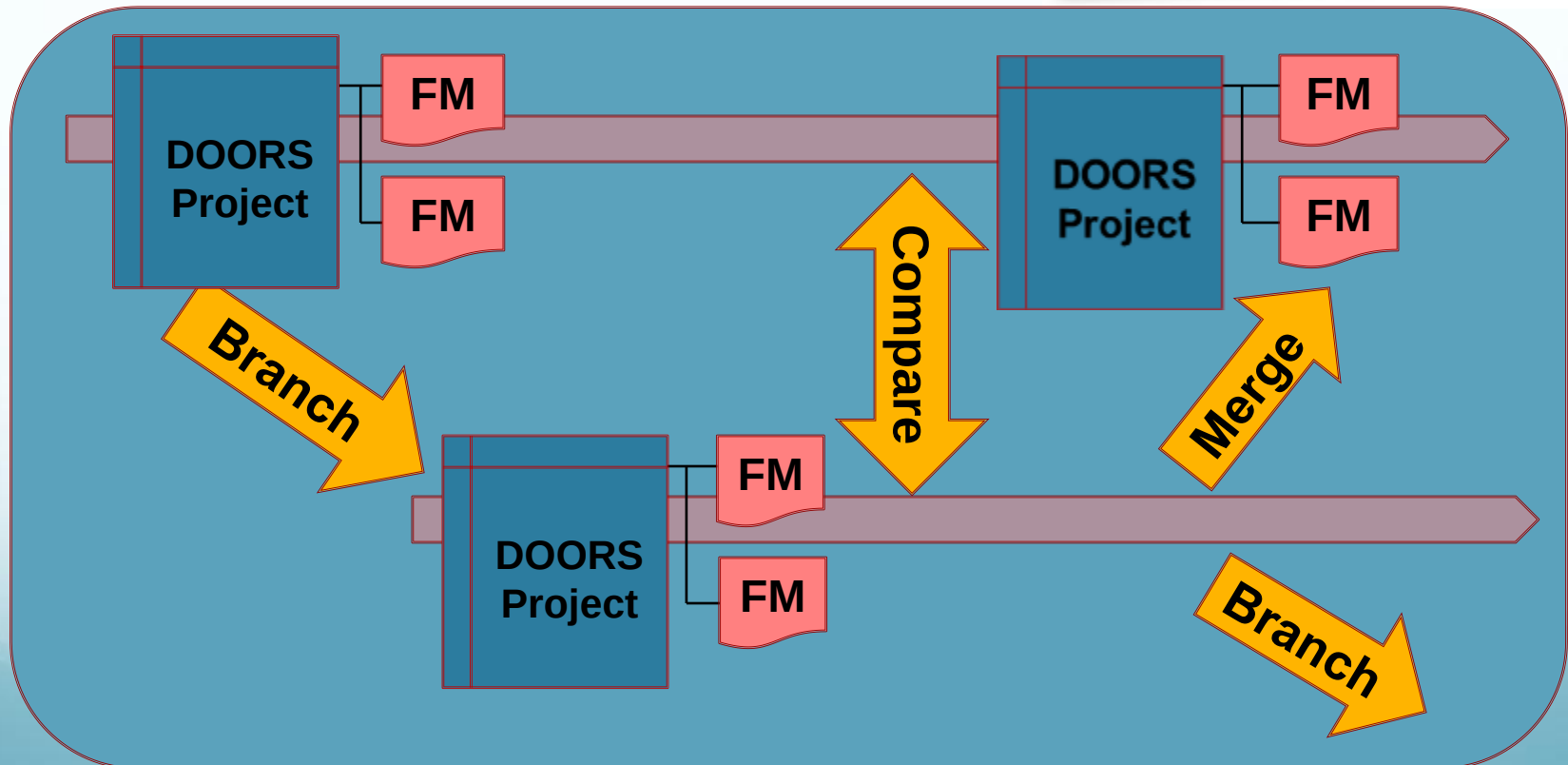
- Maintenance branching
- Personal sandbox & change evaluation
- Pull forward content



- Brings Stream Management to Rational DOORS
 - Branch sets of modules as a new stream of requirements
 - Compare modules with differences in content and structure
 - Merge changes across branches

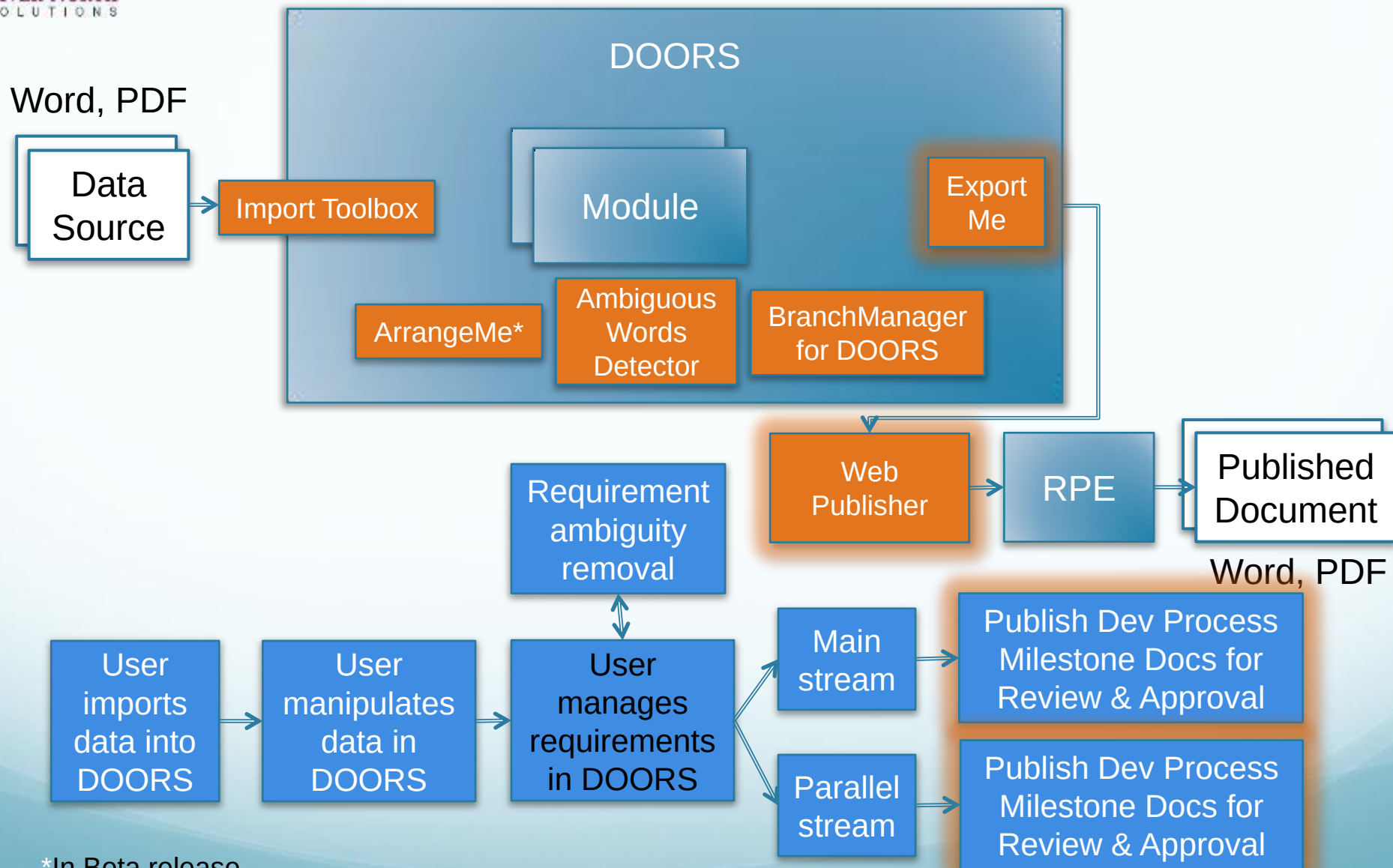
BranchManager Value Proposition

- *Product Line Engineering*
- *Maintenance Streams*
- *Requirements Library*
- *Personal Sandbox*
- ...



Accelerate Requirements Management

Workflow View



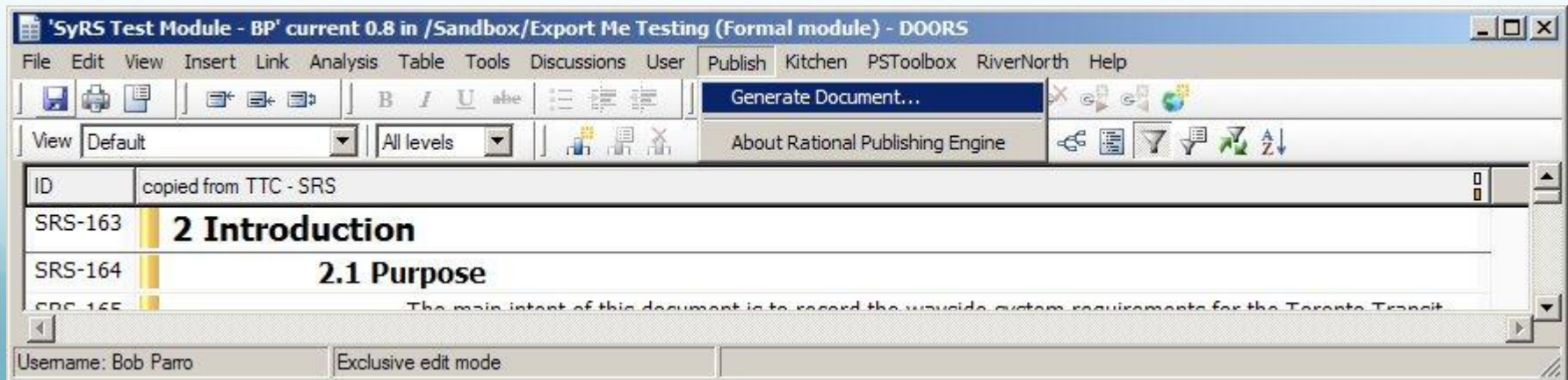
*In Beta release

Accelerate Requirements Management

Document Publishing from DOORS

■ Challenges

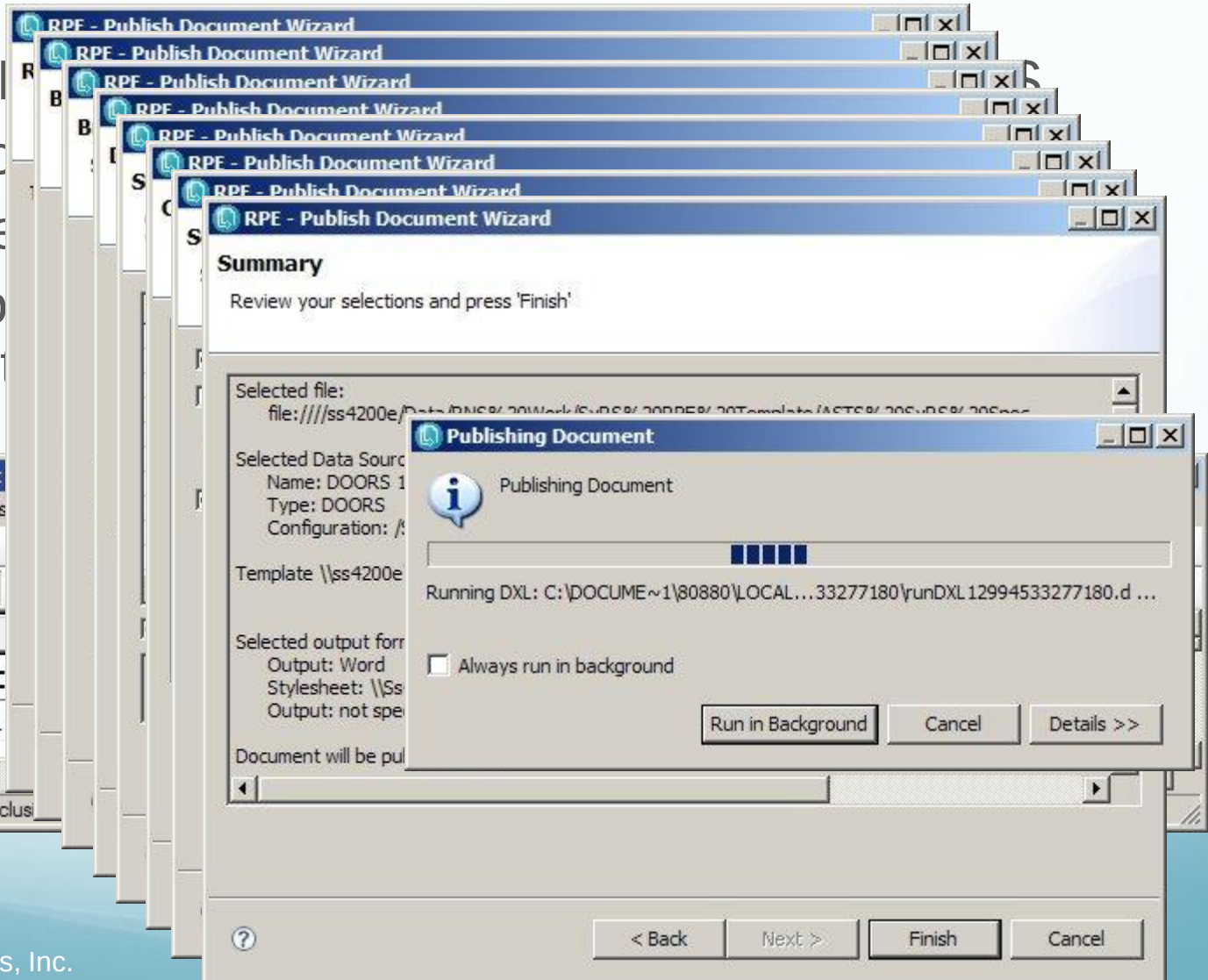
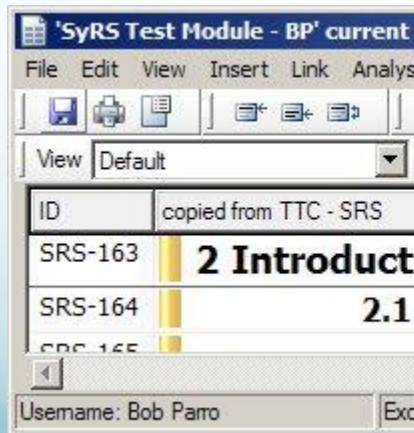
- Significant manual post-processing usually required
- Publishing document combining data from >1 module
- Template management
- RPE publish wizard in DOORS
 - Click-intensive
 - Uses terminology unfamiliar to DOORS users
 - Difficult to remember correct wizard settings to achieve standard output



Accelerate Requirements Management Document Publishing Using RPE

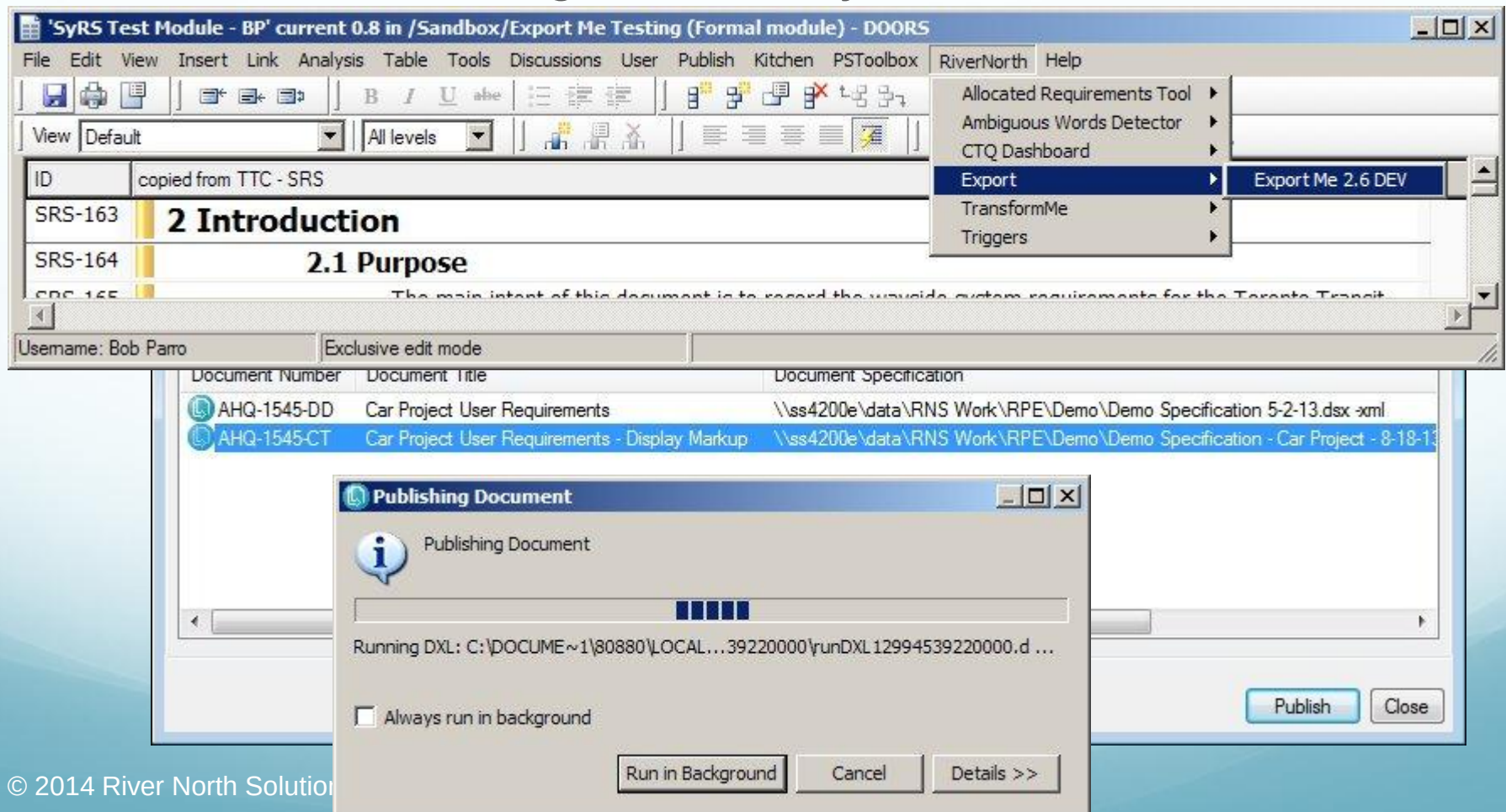
■ Challenge

- Powerful I
- Click
- Use
- Difficult to achieve s



Document Publishing...Simplified – Export Me

- Solution
 - Export Me™ single-click publishing tool for DOORS
 - Uses RPE, settings defined by RPE admin and saved



'SyRS Test Module - BP' current 0.8 in /Sandbox/Export Me Testing (Formal module) - DOORS

File Edit View Insert Link Analysis Table Tools Discussions User Publish Kitchen PSToolbox RiverNorth Help

View Default All levels

ID copied from TTC - SRS

SRS-163 **2 Introduction**

SRS-164 **2.1 Purpose**

SRS-165 The main intent of this document is to record the worldwide system requirements for the Toronto Transit

Username: Bob Parro Exclusive edit mode

Document Number Document Title Document Specification

AHQ-1545-DD Car Project User Requirements \\ss4200e\data\RNS Work\RPE\Demo\Demo Specification 5-2-13.dsx -xml

AHQ-1545-CT Car Project User Requirements - Display Markup \\ss4200e\data\RNS Work\RPE\Demo\Demo Specification - Car Project - 8-19-13

Publishing Document

Publishing Document

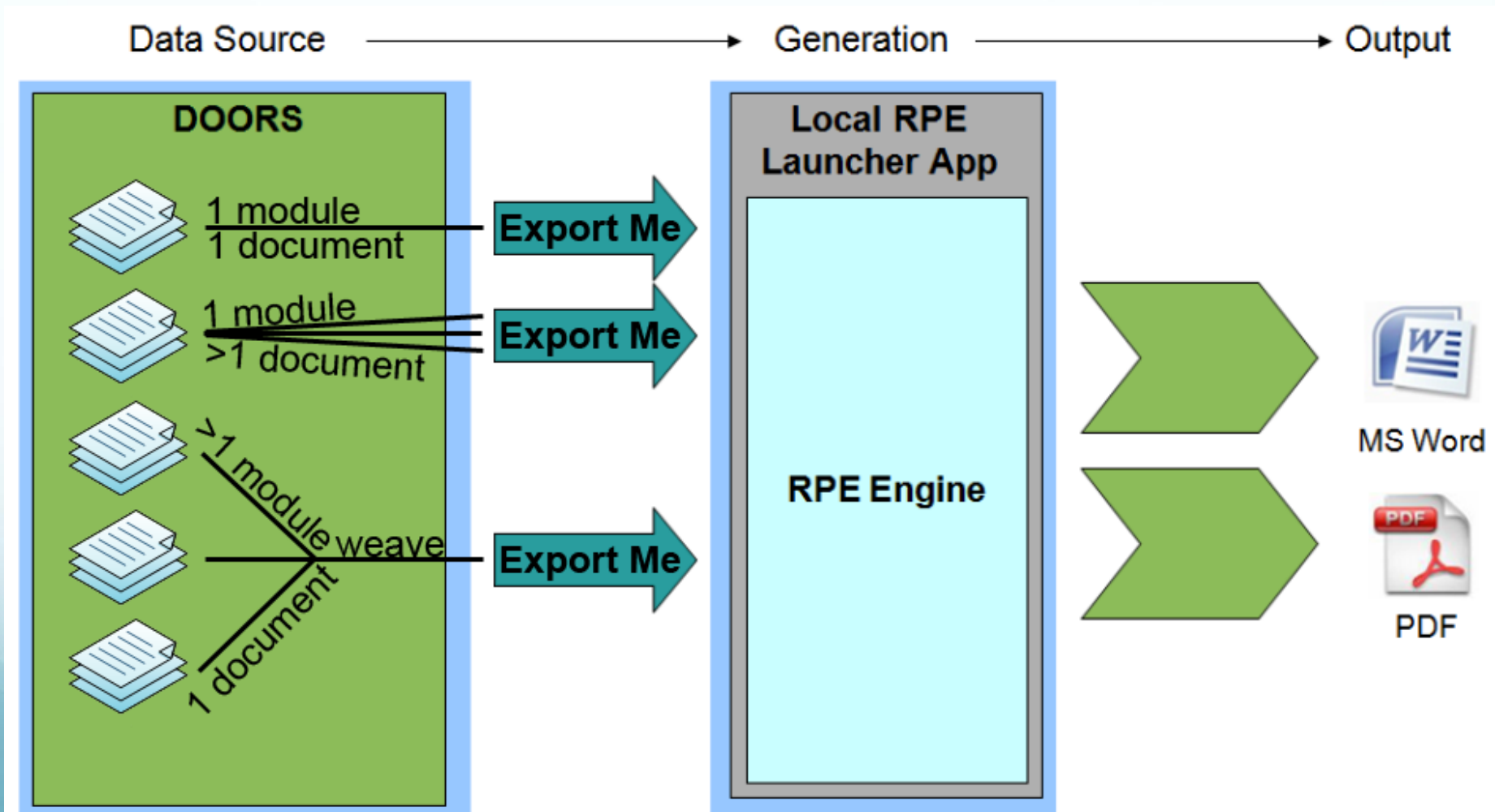
Running DXL: C:\DOCUME~1\80880\LOCAL...39220000\runDXL12994539220000.d ...

☐ Always run in background

Run in Background Cancel Details >> Publish Close

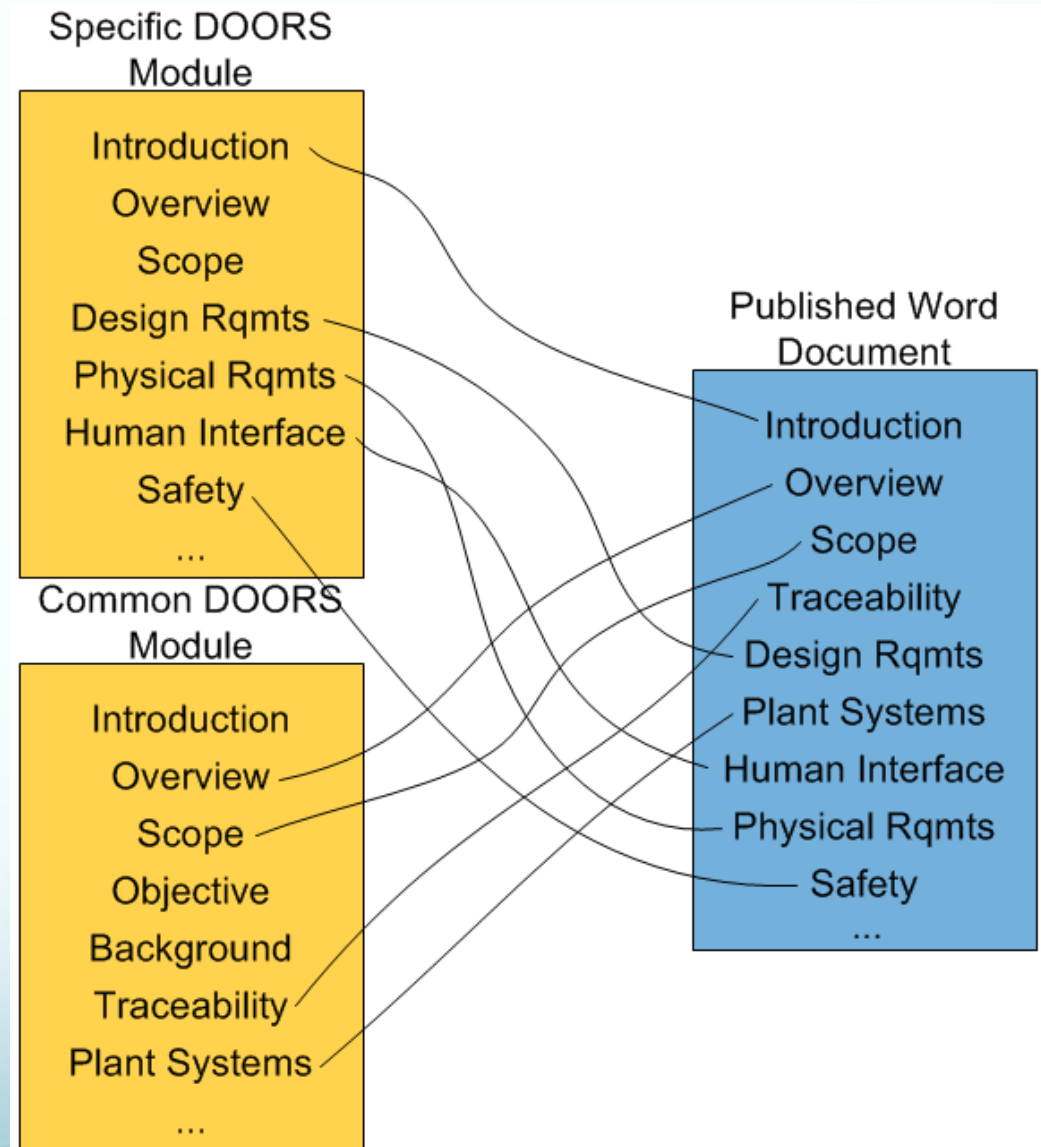
Document Publishing...Simplified – Export Me

- Export Me and Rational Publishing Engine (RPE)



Document Publishing...Simplified – Export Me

- Users want to weave data from a common data source with data from a product-specific data source
- Export Me can publish a document specification that accesses data from many data sources, in any order

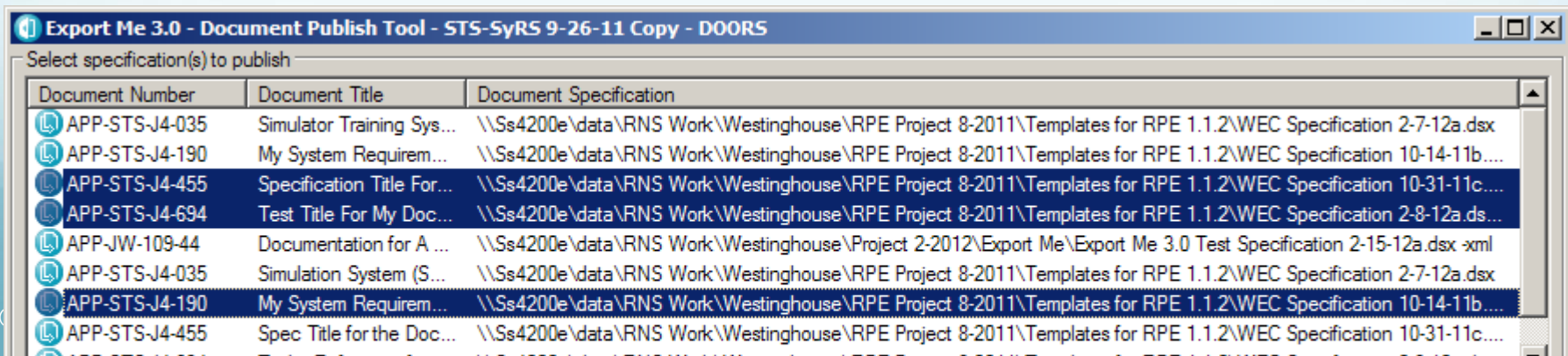


Document Publishing...Simplified – Export Me

- Users sometimes want to publish marked up version of document
- Export Me, in conjunction with add-on ‘ChangeTracker’, can optionally publish markup
- Mark up version triggered by module level attribute setting
- Useful during review/approval cycles

View Requirement Trace		All levels	
Technical Requirements		Dynamic History attribute for Object Text from 0.1 (Initial)	
☒ The Automatic Shift controller must continuously measure the time between two signal edges		The Autoshift Automatic Shift controller must continuously measures measure the time between two signal edges	
☒ 64 samples will be stored in the RAM of the controller.		If the time t > 1/baud rate, the following 64 samples will be stored in the RAM of the controller.	
☒ The ManualShift controller RF receiver will be set back to polling mode by the software controller.		The AutoShift ManualShift controller RF receiver will be set back to polling mode by the microcontroller pulling pin to '0' for a time t software controller .	

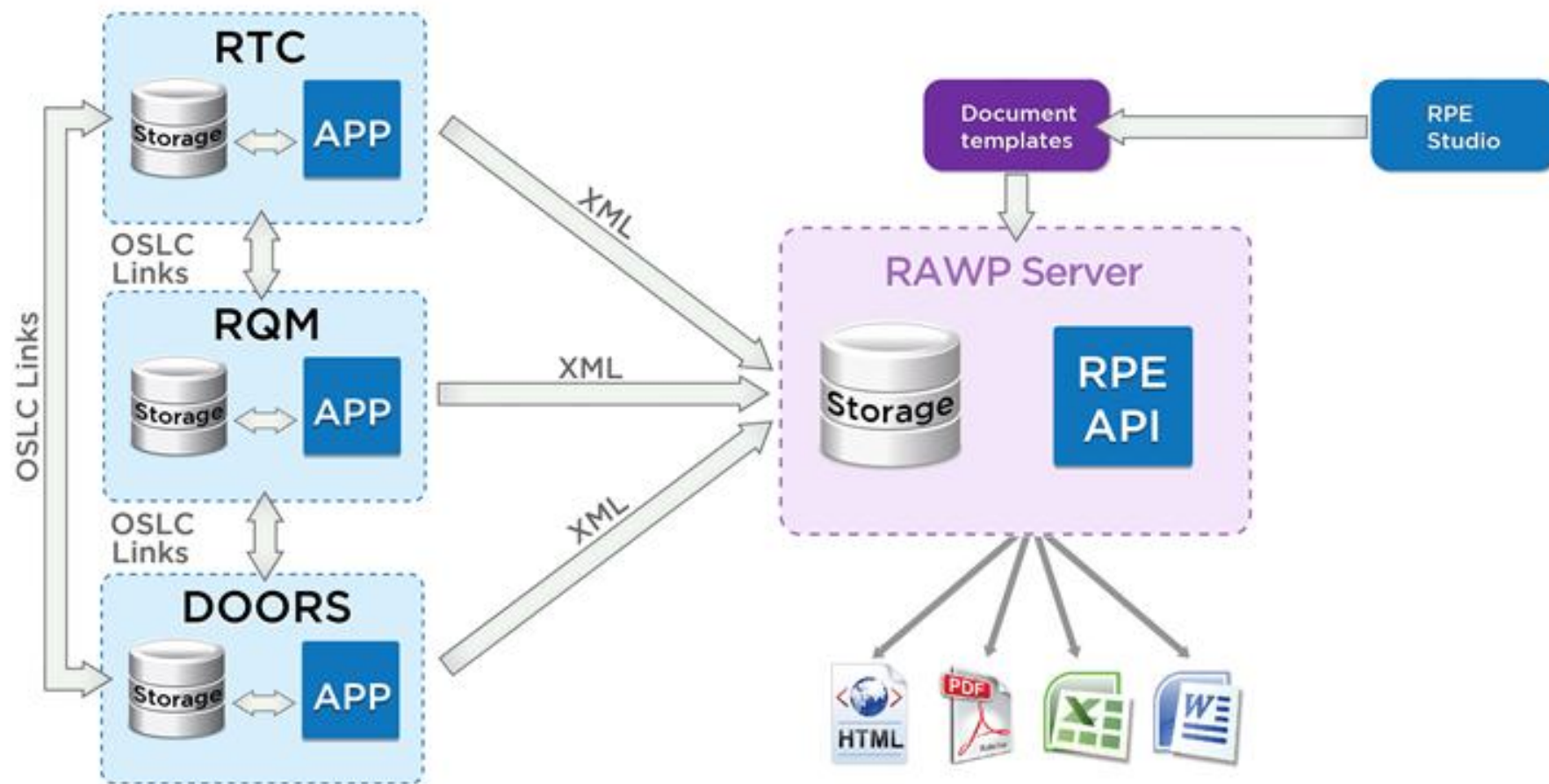
- Single-click publishing
 - No RPE/doc publishing training required for document publishers
 - Simplifies, standardizes process; opens publishing to more users; brings repeatability to the table; reduces rework
 - Publish multiple documents in one operation
 - Documentation consistency
- Publish-time auto-contextualization of RPE templates
 - Drives # RPE templates & document specifications to minimum
 - Significantly reduces template maintenance and sustainability costs – **Westinghouse: 800 -> 1**





Document Publishing Simplified – Web Publisher

- Web Publisher
 - Publish DOORS documents using web browser
 - Eliminates need to install RPE application on DOORS client machine
 - Implements RPE server
 - Facilitates sharing RPE assets across user community
 - Supports scheduling publishing tasks
 - Avoids tying up user machine to perform publishing task
 - Compatible with other tools
 - DOORS Web Access
 - Rhapsody + Design Manager
 - Rational Team Concert (RTC)
 - Rational Quality Manager (RQM)
 - XML data sources



ReportingArena
WEB PUBLISHER

dan | Help | Log Out

Specifications | Templates | Scheduler | History

My Specifications | Shared Specifications

Create... Import...

- Requirements
- Models
- Test
- Audit
- Financial
- Plan

Templates

- CLM_Traceability Report.dta
- CQ_Defects Report.dta
- ClearQuest_DefectTracking.dta
- DOORS Linked Modules.dta

Stylesheets

- rpe.dot

Document Specification - RTC - FactLab Sprint Plan v1

Start Preview Cancel

- Document Specification
 - Metadata
 - Outputs
 - Target:Html
 - Target:Word
 - Target:PDF
 - Target:XslFo
 - Templates
 - FirstPage.dta
 - RTC-Project Release Plan.dta
 - Metadata
 - Data Sources
 - Target:RTC WorkItem
 - Target:Build
 - Variables
 - Variable(hazel IRI)

Properties

Name	Value
name	baseUrl
description	
type	user
access	external
default	https://jazz.gebs.ro:9443/ccm
value	

Save Revert

Console

Starting run...

Invalid data source configuration. Not all sources are correctly configured.

Processing document template 1 of 2.

Unsigned template found. This document generation will consume a RPE license.

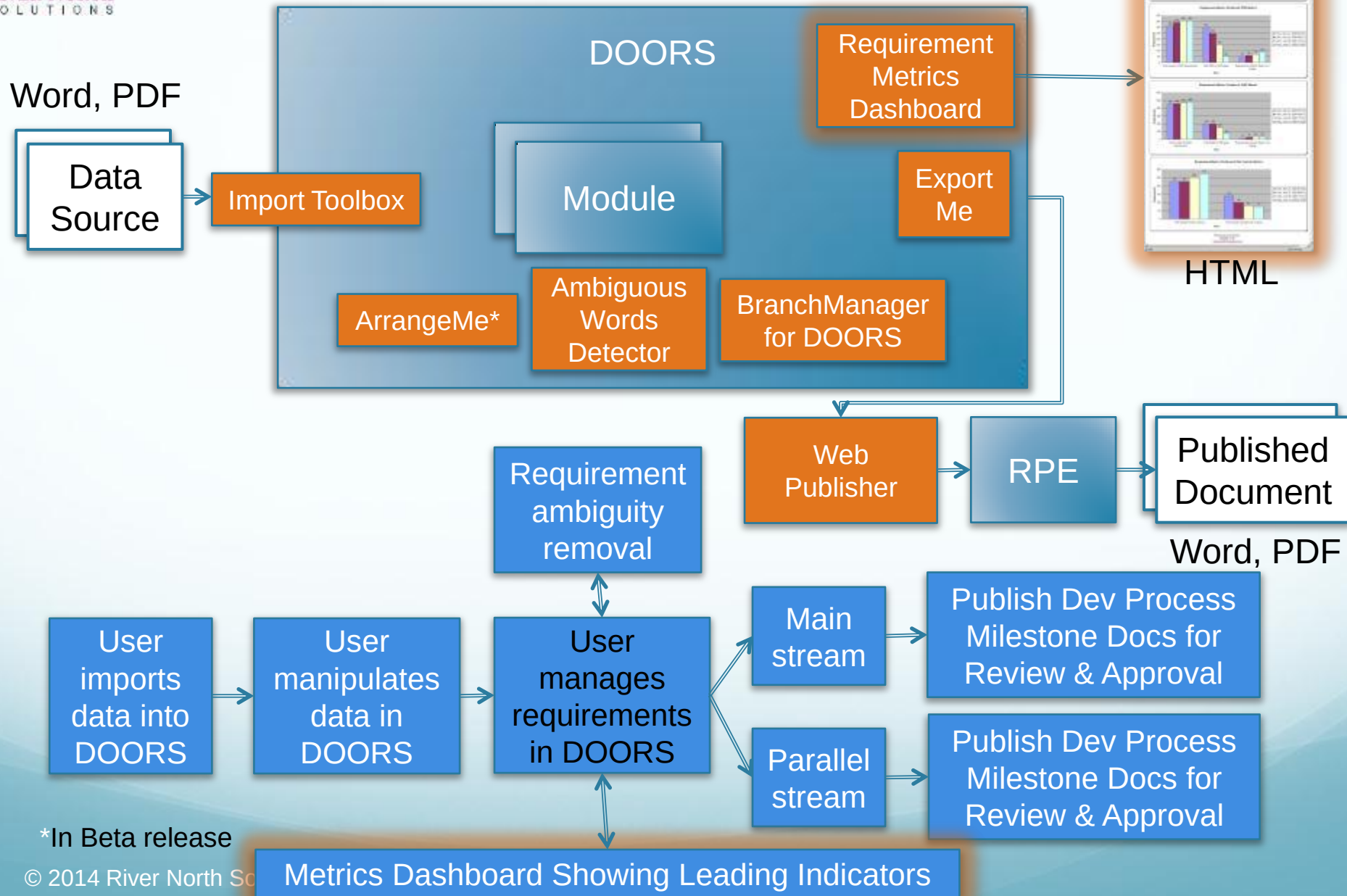
Results

Results.zip

Started Execution Time	2012-10-15 1:11:22 PM
Finished Execution Time	2012-10-15 1:11:28 PM
Execution Time	6 Seconds

- Eliminates need to install and maintain RPE application on DOORS client machine
- Frees up user machine from running publishing operation
- Schedule publishing tasks
- May allow reducing number of RPE licenses required

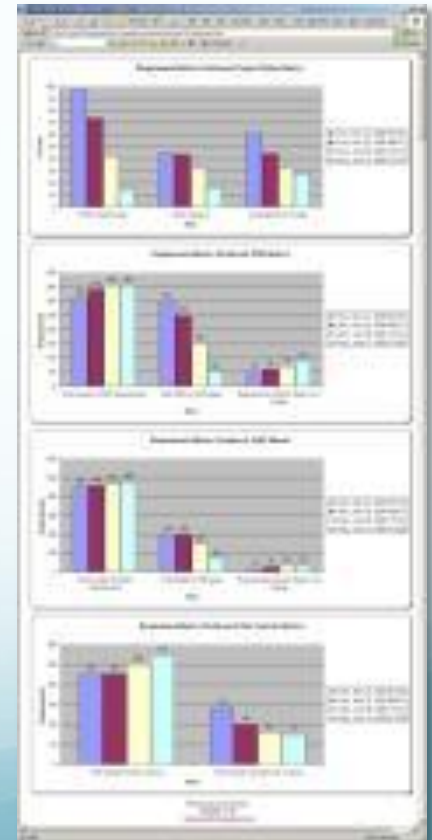
Accelerate Requirements Management Workflow View



Accelerate Requirements Management

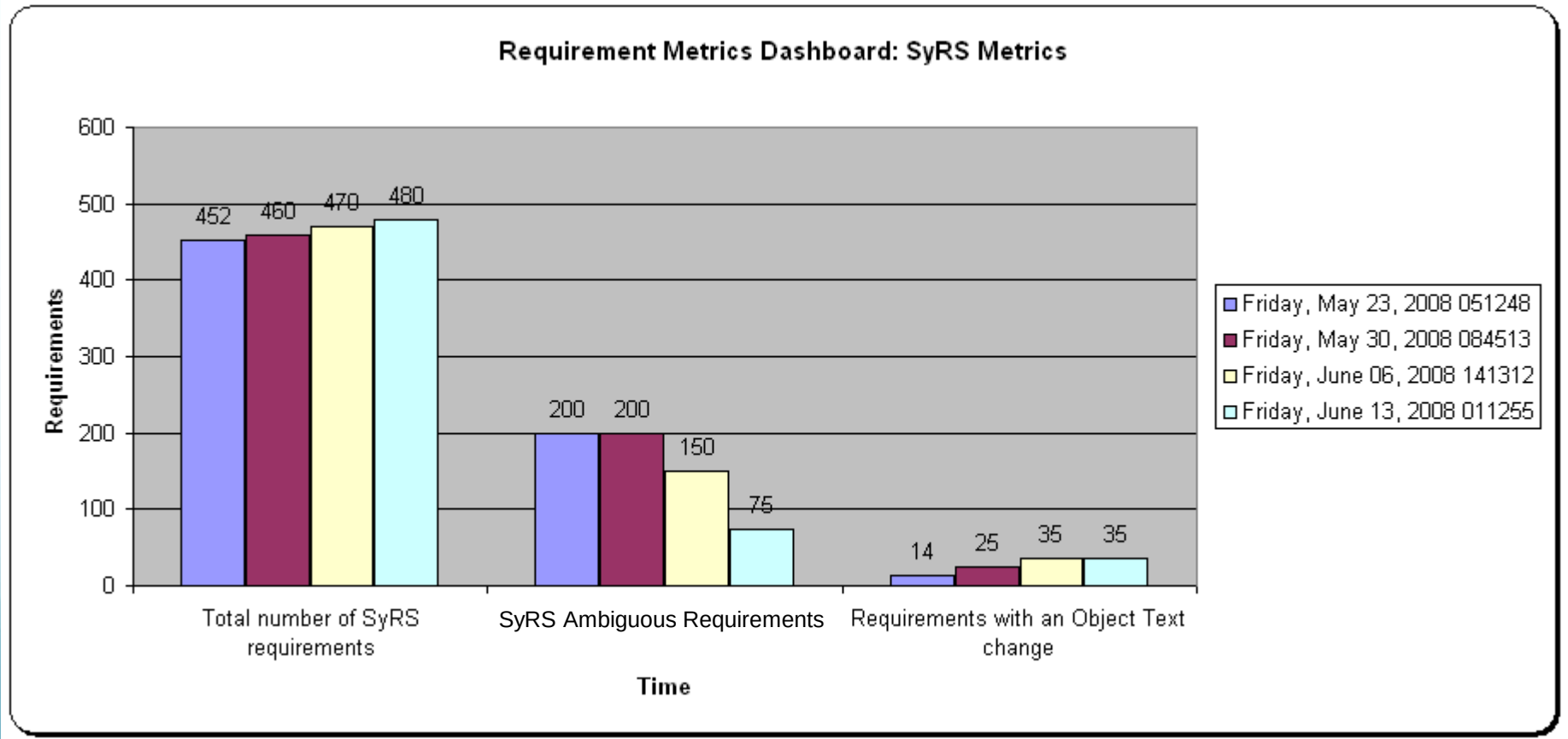
Useful for Leading Indicators

- “You can’t manage what you don’t measure”
- How do you assess risk of meeting milestones without analyzing project data and trending data?
- Early warning system – alerts risk of missing milestones
- Solution must present minimum burden on project members to obtain data



Metrics Dashboard Value Proposition

- Displays trending of project execution performance
- Easy to spot deviation from plan
- Enables corrective action well ahead of project milestones

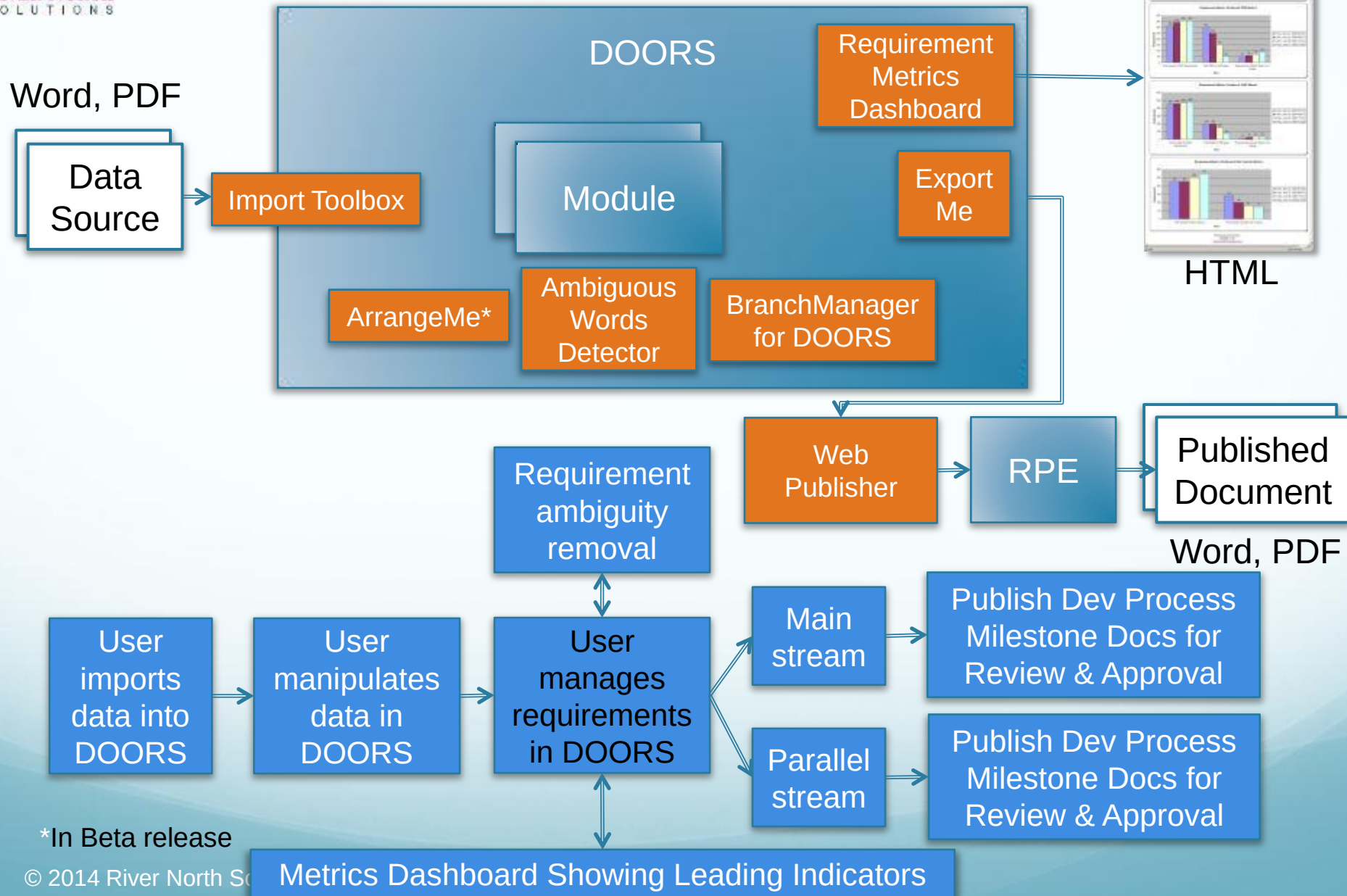


Capturing and Using Leading Indicators

- Flexible, user-defined metrics – DOORS View-based
 - View name becomes metric name
 - User creates view filter that captures metric
 - Total Rqmts, Rqmt Gaps, Rqmt Orphans, # Ambiguous Rqmts, etc.
 - Test Gaps, Test Orphans, Design Gaps, Design Orphans, etc.
 - Module name becomes metrics dashboard name
- User configures scheduler for metrics snapshot interval
- Fire and forget
 - Metrics captured at user-defined interval
 - Metrics engine builds metric charts, publishes to website
 - Runs unattended
- Metric consumers (mgrs & team members) periodically follow browser favorite to Metrics Dashboard to consume metrics

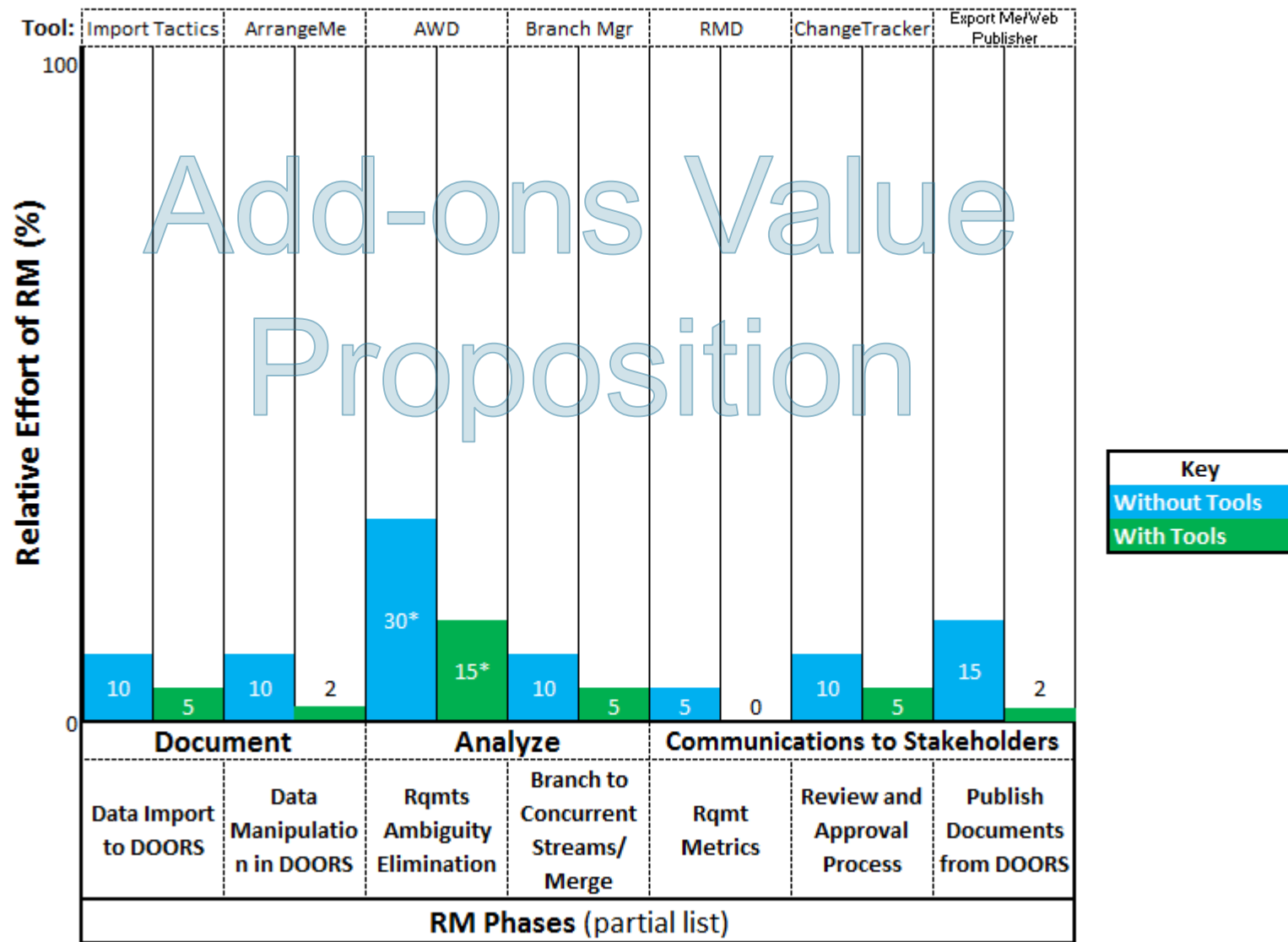
Accelerate Requirements Management

Workflow View



Accelerate Requirements Management

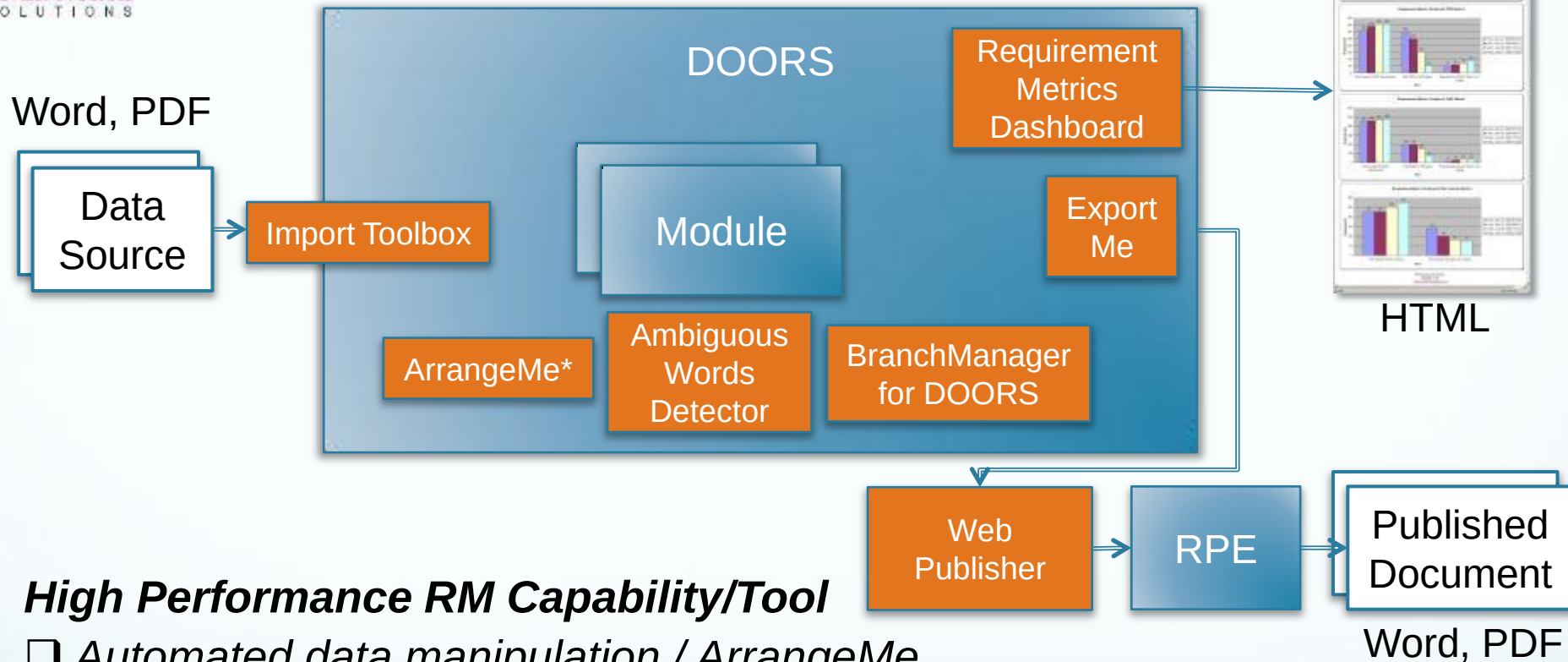
Requirements Management Effort Savings



* - reflects impact of whole development lifecycle, not just RM

Accelerate Requirements Management

Workflow View



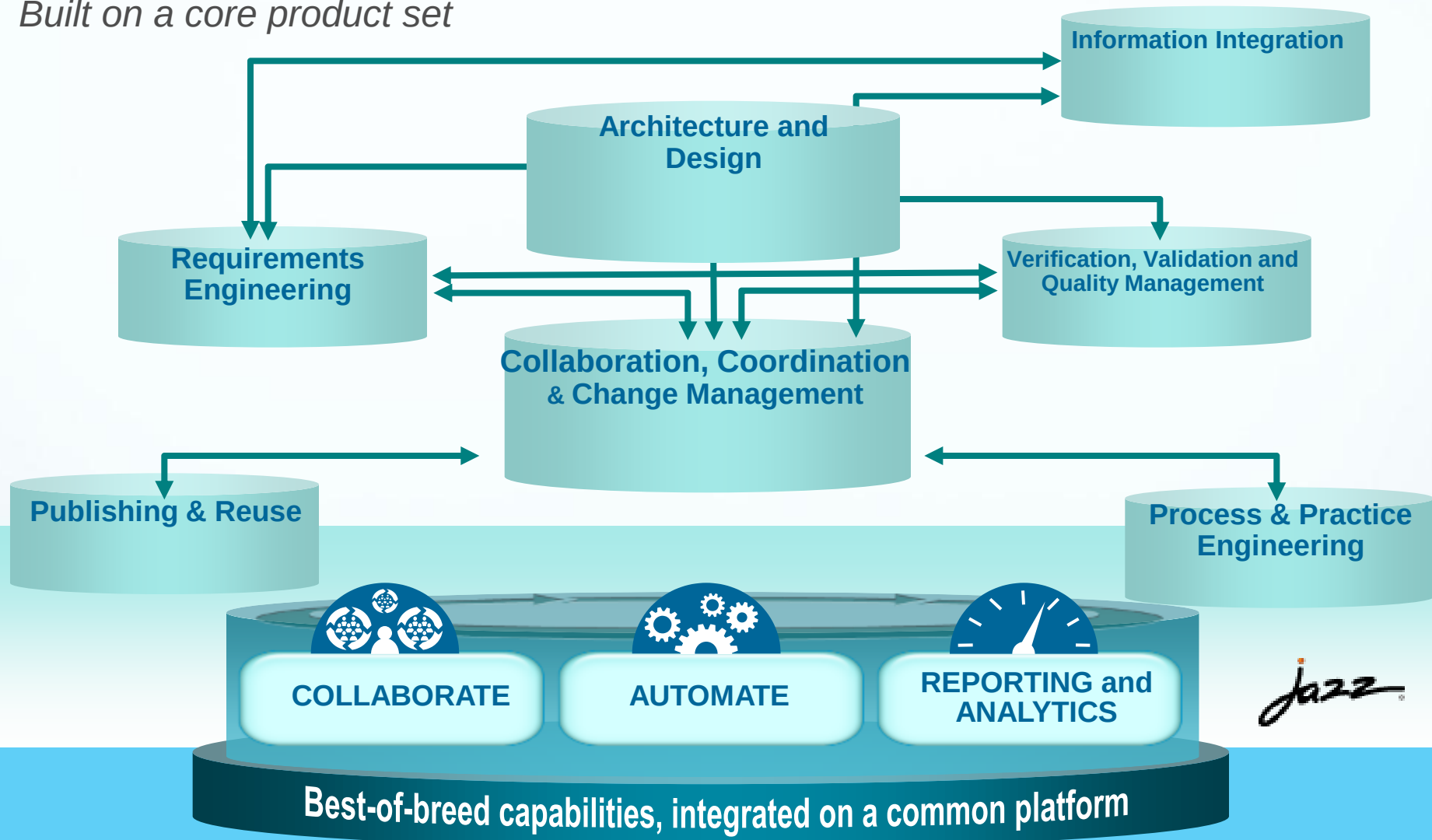
High Performance RM Capability/Tool

- ☐ Automated data manipulation / ArrangeMe
- ☐ Boost requirement quality / Ambiguous Words Detector
- ☐ Parallel stream development / BranchManager for DOORS
- ☐ Automated, consistent document publishing / Export Me
- ☐ Reduced tool footprint & licenses / Web Publisher
- ☐ Monitor leading indicator trends / Requirement Metrics Dashboard

*In Beta release

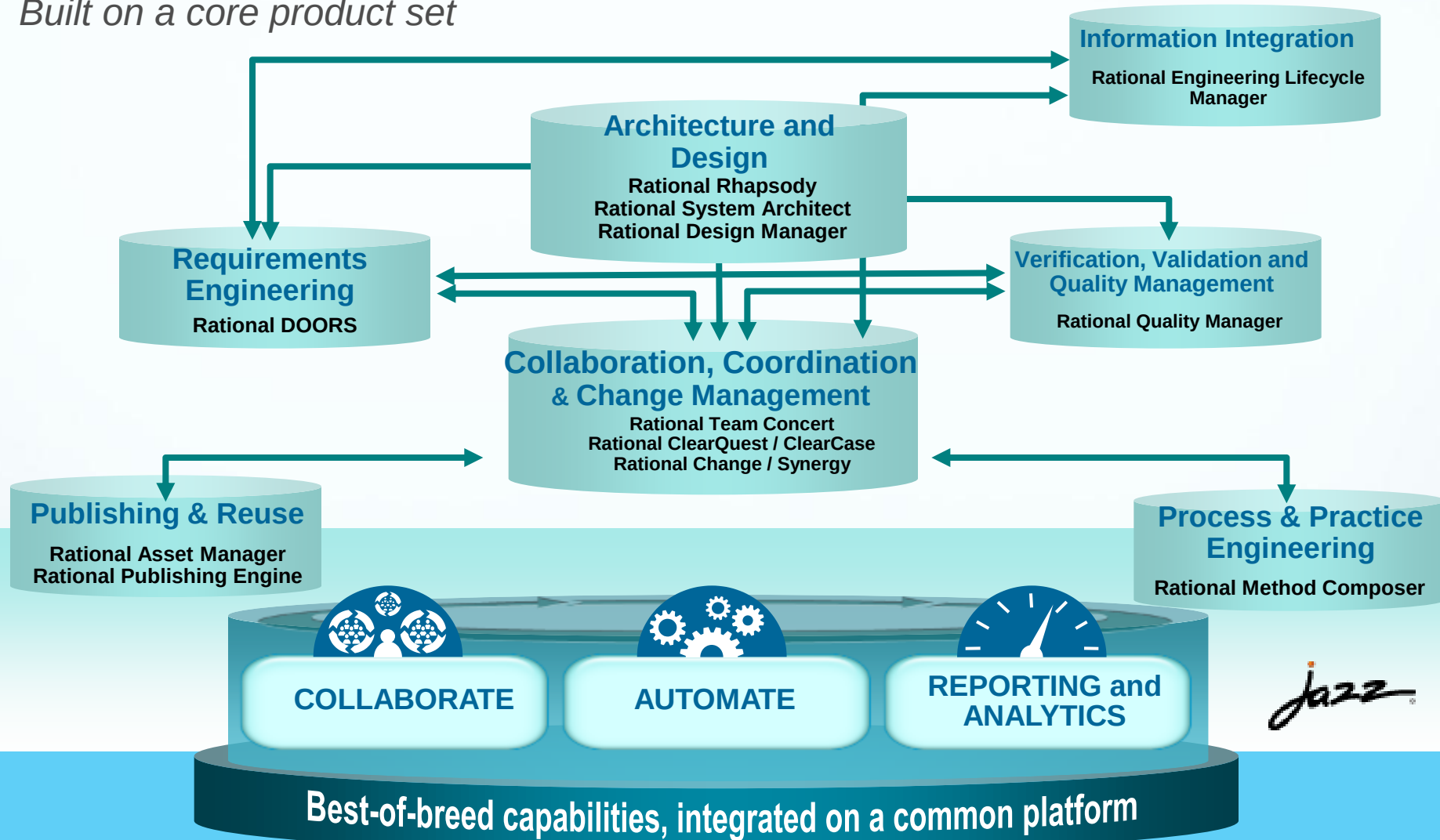
Rational Engineering Solution Platform

Built on a core product set



Rational Engineering Solution Platform

Built on a core product set



Thank You For Attending

Q&A

Bob Parro

bparro@rivernorthsolutions.com